The Metabolic Module on Betaxanthin

The offline flux store carried to the Cachera screen,

and what a physically constrained store adds

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Abstract

The metabolic module built for gene essentiality (`notes-tex/028-gene-essentiality`) carried to the target it was designed for: Cachera's betaxanthin screen, 4,930 deletion strains of BY4741 with the pigment cassette on YPD. The module is an offline store, one flux vector per deletion from a parsimonious solve on the cassette strain, read through a zero-initialized affine on the transformer's logit, and every arm is paired by seed against the same backbone without it under the 027 protocol on Merzbacher's pinned test genes. The whole-vector store lifts the width-12 backbone's pinned Spearman from 0.079 to 0.121 and, on the strain as built (the deletion-only graph), lifts the top-50 call too; the width-90 backbone without the store reads the same, and the two do not add. The gain lives in the production half of the vector and in a free affine, arrives after the backbone escapes the constant predictor, and is kept by a release rule on the validation curve. Without any backbone the clean parsimonious vector read by nearest neighbors carries 0.195, more rank than any single transformer arm, and the max-min driving-force concentrations carry 0.152 linearly. The second law as fixed reaction directions leaves the strain with no growth; as a linear program in concentrations it is consistent to within a kilojoule. Every number is at a validation-selected epoch on the pinned test set with the across-seed spread, every table is generated from the result files, and the amino-acid half of the campaign is the sibling document `notes-tex/032-amino-acid-module`.

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1 The endpoint, the protocol and the stores ✗

Cachera's betaxanthin screen measures 4,930 deletion strains of BY4741 carrying the pigment cassette on YPD; the target is the strain's betaxanthin level, and the metric the task is about is the top-50 enrichment of the highest producers beside the rank correlation. Every cell is scored under the 027 protocol: Merzbacher's 639 test genes pinned to test, the epoch selected on validation Pearson (centered five-epoch moving average), the pinned-test Spearman reported with the top-50 enrichment of high producers (a 10 percent base rate); the comparators are the 027 rounds' own numbers on the same split, read from their result files (Table 8). The backbone is the essentiality one (pathway multi-hot with ProtT5, width 12, one block, four attention heads, CRPS on the head, 200 epochs), a fifth of the width 027 trained. The graph regularizer is requested on every cell and never evaluated at width 12: it is pinned to the second block, and these cells have one, so `graph_reg_loss` is 0 at all 200 epochs. It is inert in the control and in the arm alike, so the pairing is unaffected, and every arm is paired by seed over six seeds against the same backbone with no affine.

The study began inside the essentiality campaign (rounds 117 onward there) and has been its own study, 031, since 2026-09-23, with its own results tree, W&B project and round series (B1 onward; the map from the round numbers used here to the B-series is in the study's note). The amino-acid half of the same campaign is the sibling document, `notes-tex/032-amino-acid-module`, and the essentiality endpoint the backbone and the store were built on is `notes-tex/028-gene-essentiality`.

The store represents the strain, not the wild type. For betaxanthin (`bx_reference_flux_table.py`, job 2740) the model is yeast-GEM with the cassette's 46 reactions and 20 demands, the four BY4741 markers deleted through the rules, on the corrected YPD; per GEM gene two parsimonious solves, growth and betaxanthin at 90 percent of the knockout's own growth, 8,355 columns with the total demand last; 1,155 genes, 93 lethal, 180 that move the production potential. ARO4-K229L and ARO7-G141S relieve feedback inhibition a stoichiometric model does not carry, so they change no bound. On the graph side the processor had removed the two allele genes from every Cachera cell graph, since the cassette records them as perturbations, and pooled them with the deletion; round 120's `-deletion-only-graph` keeps them in the graph and out of the pooled set, which is what the strain carries.

2 The rounds ✗

Round 117 (jobs 2741 and 2742, w134) reads the whole betaxanthin vector through the zero-initialized affine. Round 119 (jobs 2749 and 2750, w136) reads two summary columns only, the growth ratio and the production ratio, and separately the whole vector with the affine in its own optimizer group at weight decay 0.1 and a tenth of the rate. Round 120 (w137) is the deletion-only graph, and the six rule-count columns of the essentiality study's round 77 over every GEM gene. Round 122 (w139) is the affine-free control on the backbone 020 and 027 trained, width 90, nine attention heads, two blocks, where the graph regularizer is evaluated; a first build at width 48 with four heads died in that term on every cell (jobs 2757 and 2758, the round-8 failure the essentiality document records). Round 123 (w140) puts the store affine on that backbone, and on the deletion-only graph at width 12; round 125 (w142) splits the store into its growth block and its production block, 4,178 and 4,177 columns, each read alone. Round 127 (w144) adds six seeds to the best cell. Round 128 (w145) is the store on the deletion-only graph at width 90, and round 132 (w149) the wide control on that graph without the store, which completes the width-90 two-by-two of graph against store. Rounds 130 and 131 (w147, w148) answer the plateau finding of rounds 117, 123 and 125 directly: the store on the deletion-only graph with the affine's parameters frozen (`-ess-affine-start-epoch`) until epoch 40, and separately until the trailing five-epoch mean of validation Pearson reaches 0.15 (`-ess-affine-start-val-pearson`), so the backbone escapes before the affine can carry the target. A zero-initialized affine without gradient contributes exactly 0, so each cell is its control until the start epoch, which the result records. Rounds 133 to 142 screen the essentiality finalists and the physically constrained stores of Section 3, every one at width 12 on the deletion-only graph over six seeds, paired against round 120's control on that graph and against round 123's store on it: round 133 (w152) the vector with the six rule-count columns beside it, the essentiality study's best single model; round 134 (w153) the cached solve head, a hidden-256 perceptron over the store, in place of the affine; round 135 (w154) the linear minimization-of-metabolic-adjustment store; round 139 (w156) the parsimonious store rebuilt on the HiGHS two-stage path the essentiality stores use; round 140 (w157) the max-min driving-force concentration store; round 142 (w159) the mean of six samples of each deletion's flux cone. Rounds 144 and 145 (w161, w162) read the size question on the best cell, width 6 with two heads and transformer dropout 0.3 against the incumbent width 12 at 0.1. The stores are written on the target model over the source store's genes and reaction order, with the growth-ratio column unchanged, so each pair isolates the vector.

Round 117 is read (Table 1). On betaxanthin the whole-vector affine reads a pinned Spearman of 0.121 (sd 0.011 over six seeds) against the control's 0.079 (sd 0.056): +0.043 paired by seed (sd 0.049), above on five of six. What the seeds show is a change of character more than of level: the control's score depends on whether its seed leaves the plateau (0.024 to 0.174), the store arm's does not (0.102 to 0.133). Three store seeds were selected on the plateau, at validation Pearson −0.02 and epochs 127 to 181, and still read 0.12 to 0.13 on test, so the affine's linear read of the strain's flux

vector carries that rank on its own, on a test set that is 636 of 639 GEM genes. The selection rule cannot see it: the pool the validation set is drawn from holds 275 GEM genes in 4,082, and the affine reads a constant row on every other gene, so validation Pearson is nearly blind to the one term that scores on the pinned test. Averaged over the six seeds the store arm reads 0.135 (the two seed triples 0.146 and 0.133) and the control 0.076, against 027's precursor readout at 0.131 over three seeds at width 90, its context-only control at 0.115, and the cosine kNN at 0.145. On the metric the task is about, the top-50 enrichment of the highest producers, the order reverses: 3.49 for the store arm against 4.06 for the control on single seeds, 4.39 against 4.59 ensembled (a 10 percent base rate, so 35 to 46 percent precision). The vector lifts the rank correlation of a small backbone to where 027's larger one sat and does not lift its top-50 call.

Rounds 119 and 120 are read, and they locate the round-117 gain in the whole vector read through a free affine. The two summary columns read 0.101 (sd 0.061) against the control's 0.079, +0.023 paired (sd 0.072), above on four of six: three seeds were selected at epoch 197 at validation Pearson 0.04 to 0.05 and read 0.13 on test, the round-117 pattern, and one seed reads −0.021, so the level comes with the control's spread rather than the whole vector's. Ensembled over six seeds it reads 0.071, under the control's 0.076, and its single-seed top-50 enrichment is 3.03. The whole vector with the affine decayed and slowed reads 0.073 (sd 0.017), −0.005 paired (sd 0.058), above on three of six; three of its seeds were selected at epoch 2 at validation Pearson 0, where the affine has taken two steps, and read 0.05 to 0.08. So the rate the affine trains at is part of what round 117 measured, and slowing it returns the arm to the control. The deletion-only graph without an affine reads 0.058 (sd 0.075), −0.020 paired (sd 0.076), above on two of six, with one seed at −0.089 and every selection between epochs 3 and 69; the six rule-count columns read 0.074 (sd 0.059), −0.004 paired (sd 0.092), above on two of six. Neither the graph nor the rule counts moves the rank, in either direction, beyond the seed spread. One read cuts across all four: the six-seed ensemble's top-50 enrichment is 4.8 to 5.2 for every arm of rounds 119 and 120 against 3.0 to 3.6 for their single seeds, and 4.59 for the control's ensemble, so on the call the task is about, averaging seeds is worth more than any of these representations, as it was on the essentiality endpoint (the essentiality document's fold-ensemble section). Each ensemble is one number with no spread, so that ordering among the arms is not a measurement.

The deletion-only half of round 123 is read (job 2760, 3 h 34 min for six cells at a 32.6 GB peak): the store affine on the graph that carries the two alleles, the strain as built. Against the deletion-only graph without the affine it reads 0.121 (sd 0.025) against 0.058, +0.063 paired (sd 0.065), above on six of six; against the round-117 store arm on the old graph it reads −0.000 paired (sd 0.017), above on four of six, so the rank the affine carries is the same on either graph. What changes is where the seeds are selected and what the call is worth. Five of six were selected off the plateau, at validation Pearson 0.20 to 0.27 and epochs 46 to 105, where round 117's seeds sat on it at −0.02; the sixth (seed 2704) was selected on the plateau at epoch 149 and its pinned predictions are rank-identical to round 117's seed 2704 (Spearman 1.0 between the two arms, largest difference 0.004), which is what a plateau cell is: the affine's read of the store and nothing from the graph. The top-50 enrichment is 4.93 on single seeds against 3.49 for the store on the old graph, 3.39 for the deletion-only graph alone and 4.06 for the control, and 5.59 ensembled over six seeds, the highest of any arm; the six-seed rank is 0.131. So the strain represented as built is the first cell where the store lifts both the rank and the call, and it is the arm the width and block rounds are now measured against.

The betaxanthin half of round 122 is read (jobs 2762 and 2763, 7 h 11 min and 7 h 23 min for three cells each at a 16.5 GB peak): the affine-free control on 027's backbone, width 90, nine heads, two blocks, on the old graph. It reads 0.109 (sd 0.053) against the width-12 control's 0.079, +0.030 paired (sd 0.079), above on four of six, with a top-50 enrichment of 4.73 against 4.06; ensembled over six seeds 0.134 and 5.59, where 027's own three-seed cell of this form read 0.115. Every wide seed leaves the plateau, selected at epochs 9 to 107 with validation Pearson 0.16 to 0.36, where the width-12 arms lose a third of their seeds to it; the validation loss bottoms at a median epoch of 42 and then rises by 0.01 to 0.11 to epoch 199, and two seeds fall back to a constant predictor by the end. So width buys the escape and the early peak, and 200 epochs is four times what these cells use. Against the width-12 store affine on the corrected graph (0.121, 4.93, ensembled 0.131 and 5.59), the seven-times-wider backbone without the store reads the same on every measure, which is the size of the store's contribution at width 12. Whether the two add is round 128 (w145), the store on the corrected graph at width 90.

The wide half of round 123 is read (jobs 2766 and 2767, 7 h 25 min and 7 h 20 min): the store affine on 027's backbone, old graph. It reads 0.134 (sd 0.020) against the wide control's 0.109, +0.025 paired (sd 0.051), above on four of six, and ensembled over six seeds 0.153, the highest rank of any cell on this split, above 027's precursor readout at 0.131 and the cosine kNN at 0.145. The call moves the other way, as in round 117: top-50 enrichment 4.09 against the wide control's 4.73 on single seeds and 5.19 against 5.59 ensembled. The seeds show why. Four of the six were selected on the plateau, at validation Pearson −0.06 to 0.06 and epochs 9 to 196, where the wide control had no plateau seed at all; the two that escaped were selected at 0.29 and 0.30. So at width 90 the affine's read of the store carries the rank on its own while the backbone stays constant on most seeds, and the top-50 call, which the escaped backbone supplies, follows the escape rate. On the old graph the width-12 store had three plateau seeds of six and the corrected graph one; round 128 puts the store on the corrected graph at width 90.

Round 125 is read (jobs 2768 and 2769, 4 h 29 min and 4 h 38 min): the whole vector's two blocks read alone, at width 12 on the old graph, each paired against the whole vector of round 117 and against the control. The growth block

alone, 4,178 columns, reads 0.033 (sd 0.021): −0.088 paired against the whole vector (sd 0.029), above on no seed, and −0.046 against the control (sd 0.064), above on one of six, with a top-50 enrichment of 3.06; ensembled 0.059. The production block alone, 4,177 columns, reads 0.091 (sd 0.040): −0.030 against the whole vector (sd 0.034), above on one of six, and +0.013 against the control (sd 0.021), above on five of six, top-50 3.63; ensembled 0.112. So the rank the affine carries lives in the production half, the parsimonious solve at 90 percent of the knockout's own growth with the betaxanthin demand as the objective, and the growth half carries none of it on its own and costs the control 0.046 when read alone; yet the whole vector beats the production half by 0.030 on five of six seeds, so the growth columns add to the production columns what they subtract from the backbone. Three seeds of each half were selected on the plateau, as with the whole vector, and the validation loss bottoms at a median epoch of 53 on both.

Round 127 is read (job 2771, 3 h 40 min): six more seeds, 2706 to 2711, of the store on the corrected graph, so the best betaxanthin cell has twelve. The new six read 0.058 to 0.138, mean 0.111, every one selected off the plateau at validation Pearson 0.15 to 0.22, so over twelve seeds the arm reads 0.116 (sd 0.027, standard error 0.008) with a top-50 enrichment of 4.86. Averaging all twelve reads 0.129 and 5.39 against 0.131 and 5.59 for the first six, so the ensemble has saturated at six on this cell: the second six seeds add nothing to the rank or the call, and the ensemble numbers of the transfer arms, one draw each, are read as sitting within about 0.02 and 0.2 of where more seeds would put them.

Round 128 is read (jobs 2772 and 2773, 7 h 04 min and 7 h 31 min for three cells each): the store affine on 027's backbone, width 90, on the deletion-only graph, the fourth cell of the width-90 two-by-two. It reads 0.120 (sd 0.024) against the wide control on the old graph at 0.109, +0.011 paired (sd 0.067), above on three of six, and against the width-12 store on the same graph at 0.121, −0.001 paired (sd 0.035), above on three of six; top-50 enrichment 4.43 against 4.73 and 4.93. Ensembled over six seeds it reads 0.147 and 4.99, under the old-graph wide store's 0.153 and the width-12 corrected-graph store's 5.59. What the graph correction changes at width 90 is the plateau, as it did at width 12: every seed is selected off it, at epochs 21 to 50 with validation Pearson 0.22 to 0.28, where the old-graph wide store had four plateau seeds of six. The rank does not follow. Width buys the escape (round 122), the store buys the rank at width 12 (round 123), and put together on the strain as built they read what either reads alone, so the two contributions do not add, and the seed spread (sd 0.024) is the smallest of the four wide cells. The validation loss bottoms at epochs 8 to 42 and rises 0.09 to 0.15 by epoch 199 on every seed, the round-122 pattern, so these seven-hour cells spend six hours past their minimum. Round 132 (w149), the wide control on this graph, completes the square.

Rounds 130 and 131 are read (jobs 2782 and 2783, 3 h 52 min and 3 h 46 min), the two schedules for the store affine on the deletion-only graph. Round 130 holds the affine at zero until epoch 40: it reads 0.074 (sd 0.046) against the store arm's 0.121 (−0.047 paired, sd 0.053, above on two of six) and against the affine-free control's 0.058 (+0.016, sd 0.068, four of six), top-50 3.86 against 4.93 and 3.39, and ensembled over six seeds 0.129 with a top-50 of 5.79 against 0.131 and 5.59. The affine started at epoch 40 in every seed, and five of the six seeds select their epoch before it (10, 32, 34, 25 and 38; the sixth at 76), so the selected cell is the affine-free trajectory in five seeds of six: the validation peak the affine-free model reaches before epoch 40 is not overtaken once the affine arrives. Round 131 releases the affine from epoch 5 once the trailing validation Pearson reaches 0.15: the release came at epochs 5, 6, 25, 31, 6 and 40, and the seeds select at 2, 57, 35, 32, 78 and 38, four of them after the release. It reads 0.110 (sd 0.050), −0.010 against the store arm (sd 0.052, one of six) and +0.052 against the control (sd 0.035, six of six), top-50 4.66, ensembled 0.120 with a top-50 of 6.19, the highest ensemble call of any betaxanthin arm. The seed whose release fell at epoch 40 is the same trajectory in both rounds (0.016 at epoch 38 in each), a check that the schedule and not the seed is what separates them. So the store's gain does not come from the first epochs: the affine released after the model leaves the constant predictor reads the store arm's level and its best call, and the affine held to epoch 40 loses the level because the selected epoch comes first. A release rule on the validation curve costs nothing and keeps the gain; a fixed late start does not.

Round 132 is read (jobs 2784 and 2785, 7 h 18 min and 7 h 08 min for three cells each): the width-90 backbone on the deletion-only graph without the store, the last cell of the width-90 two-by-two. It reads 0.092 (sd 0.043) against the old-graph wide control's 0.109 (−0.017 paired, sd 0.080, above on two of six), top-50 4.49 against 4.73, and ensembled over six seeds 0.122 with a top-50 of 5.19 against 0.134 and 5.59. Every seed is selected off the plateau, at epochs 19 to 56 with validation Pearson 0.22 to 0.29, and the validation loss bottoms at epochs 26 to 53 and rises by 0.08 to 0.17 by epoch 199, the largest rise of any arm. The square is now paired on its second diagonal: the store on this graph at width 90 (round 128) against this control reads +0.028 (sd 0.037, above on four of six), with top-50 4.43 against 4.49. So at width 90 the graph correction takes 0.017 off the control and the store puts 0.028 back, the store's contribution on the strain as built is the same 0.03 at width 90 as it is at width 12 against the old-graph control, and none of the four wide cells beats the width-12 store on the same graph (0.121, top-50 4.93). Width costs seven hours a cell and buys nothing the width-12 store does not have.

Round 134 is read (job 2788, 3 h 42 min): the cached solve head, a hidden layer of 256 over the store vector in place of the zero-initialized affine, on the deletion-only graph. It reads 0.070 (sd 0.049) against the affine-free control's 0.058 (+0.012 paired, sd 0.074, above on three of six) and against the store affine's 0.121 (−0.051, sd 0.061, one of six), top-50 3.89 against 3.39 and 4.93, and ensembled over six seeds 0.095 with a top-50 of 5.39 against 0.082 and 0.131. Two seeds never leave the constant predictor: their validation Pearson peaks at −0.024 and 0.003 over the 200 epochs, they are selected at epoch 2 and read 0.038 and 0.005. The four that escape are selected at epochs 43 to 94 with validation

Pearson 0.18 to 0.26 and read 0.043 to 0.136, at the affine's level on two seeds and under it on two. So the head that recovered the affine's loss on the amino acids (the sibling document, round 137) loses the affine's gain here and brings back the plateau the deletion-only graph had removed: a zero-initialized affine contributes 0 until it learns and cannot hold the backbone down, and a randomly initialized perceptron over 8,355 columns can. The linear read of the store is the one that travels.

Round 135 is read (job 2792, 3 h 37 min): the linear minimization-of-metabolic-adjustment store, the nearest feasible flux to the parent's parsimonious flux under each knockout, in the affine slot on the deletion-only graph. It reads 0.107 (sd 0.018) against the affine-free control's 0.058 (+0.049 paired, sd 0.064, above on five of six) and against the parsimonious store's 0.121 (−0.014, sd 0.029, one of six), with a top-50 enrichment of 4.99 against 3.39 and 4.93, the highest single-seed call of any arm, and ensembled over six seeds 0.115 with a top-50 of 5.79 against 0.082 and 0.131. Five seeds are selected off the plateau at epochs 30 to 59 with validation Pearson 0.22 to 0.26; the sixth is selected on it at epoch 3 and still reads 0.098, the affine's read of the store alone, as in round 117. So a store anchored to one wild-type vector, which Section 3 measures as almost free of the vertex degeneracy (18 no-effect rows differ from the wild type's against the parsimonious store's 1,025), reads the parsimonious store's rank to within the seed spread with half its spread and a better call. The adjustment program is a second physically stated store that carries the target in the transformer; whether the degeneracy itself costs rank is round 139, the parsimonious store on one solver path.

Round 133 is read (job 2789, 3 h 45 min): the parsimonious vector with the six rule-count columns beside it, the essentiality study's best single store, in the affine slot on the deletion-only graph. It reads 0.114 (sd 0.019) against the affine-free control's 0.058 (+0.056 paired, sd 0.056, above on five of six) and against the plain store's 0.121 (−0.007, sd 0.023, one of six), top-50 3.96 against 3.39 and 4.93, and ensembled over six seeds 0.137 with a top-50 of 4.79 against 0.082 and 0.131, the highest six-seed rank of any width-12 arm. Five seeds are selected off the plateau at epochs 47 to 91 with validation Pearson 0.18 to 0.27; the sixth sits on it to epoch 183 and reads 0.118, the store's rank without the backbone. So the six columns that lifted the essentiality score keep the store's rank here, cost its single-seed call a point of enrichment, and add to the seed ensemble what they take from the single seed. On the amino acids the same columns read below both controls (the sibling document, round 136).

Round 139 is read (job 2794, 3 h 39 min): the parsimonious store rebuilt on the HiGHS two-stage path the essentiality stores use, the vector without the GLPK build's vertex noise (Section 3: 428 no-effect rows differ from the wild type's against 1,025), in the affine slot on the deletion-only graph. It reads 0.130 (sd 0.034) against the affine-free control's 0.058 (+0.072 paired, sd 0.067, above on five of six) and against the GLPK store's 0.121 (+0.009, sd 0.041, five of six), top-50 4.46 against 3.39 and 4.93, and ensembled over six seeds 0.148 with a top-50 of 5.79 against 0.082 and 0.131, the highest six-seed rank of any width-12 arm and within 0.005 of the width-90 store's 0.153. Five seeds select off the plateau at epochs 30 to 94 with validation Pearson 0.21 to 0.26 and read 0.076 to 0.149; the sixth is selected on it at epoch 2 with validation Pearson 0.02 and reads 0.160, the highest single cell of any arm, the affine's linear read of the clean vector with nothing from the backbone (the GLPK store's plateau cells read 0.12 to 0.13). So the vertex noise costs the store about 0.01 of rank on five seeds of six and 0.02 on the ensemble, the clean vector read alone carries more than the GLPK vector read alone, and the hypothesis of Section 3 reads as measured in that direction and small: one solver path is the right build, and it is not what separates the store from the width-90 ensemble.

Round 140 is read (job 2795, 3 h 35 min): the metabolite side in the affine slot, the max-min driving-force concentrations of Section 3 (2,831 metabolites in mM with the driving force, the count of constrained reactions and the count binding, under the 100 kJ/mol cap), on the deletion-only graph. It reads 0.118 (sd 0.055) against the affine-free control's 0.058 (+0.059 paired, sd 0.055, above on five of six) and against the GLPK flux store's 0.121 (−0.003, sd 0.037, two of six), top-50 4.06 against 3.39 and 4.93, and ensembled over six seeds 0.133 with a top-50 of 4.99 against 0.082 and 0.131. The seeds spread more than the flux store's: two read 0.167, selected at epochs 23 and 6 (the second on validation Pearson 0.13, before the backbone has escaped), and two read 0.058 and 0.040, selected at 106 and 50 after the validation loss had risen. So the concentration vector, a different physical quantity from the flux vector and derived from it by a program that adds the second law, carries the same rank in the transformer as the flux vector it is built on, with twice the spread and a weaker call; the ladder read it the same way (0.152 by ridge against the HiGHS store's 0.195 by nearest neighbors). What the second law adds on this target is a second representation at the flux store's level, not a higher one.

Round 142 is read (job 2797, 3 h 39 min): the sampled cone in the affine slot, the mean of six OptGP draws of each deletion's flux cone at 99 percent of its optimum (Section 3), the Flux Cone Learning representation on this backbone, on the deletion-only graph. It reads 0.088 (sd 0.022) against the affine-free control's 0.058 (+0.030 paired, sd 0.070, above on four of six) and against the GLPK parsimonious store's 0.121 (−0.033, sd 0.024, above on no seed), top-50 2.90 against 3.39 and 4.93, and ensembled over six seeds 0.083 with a top-50 of 3.00 against 0.082 and 0.131. Every seed selects off the plateau, at epochs 35 to 138 with validation Pearson 0.11 to 0.25, and reads 0.072 to 0.131. So the cone's mean carries less rank than the optimum it is sampled around, on every seed, and takes the call below the control's, the same order the ladder read without a backbone (0.007 by ridge, 0.064 by nearest neighbors against the point's 0.195): on this target a deletion's feasible region averaged over six draws is a worse representation than one optimum inside it, as the sampled stores were on the essentiality endpoint (the essentiality document, round 87), and the representation

the Flux Cone Learning comparison rests on does not travel to the transformer.

Round 144 is read (job 2798, 3 h 12 min): the size question from the small side, the store arm on the deletion-only graph at width 6 with two attention heads, half the incumbent's width, paired against the width-12 store. It reads 0.113 (sd 0.038) against 0.121 (−0.008 paired, sd 0.055, above on four of six), top-50 4.19 against 4.93, and ensembled over six seeds 0.129 with a top-50 of 5.39 against 0.131 and 5.59. Four seeds select off the plateau at epochs 39 to 86 with validation Pearson 0.20 to 0.25 and read 0.110 to 0.137; two never leave it, selected at epochs 185 and 197 on validation Pearson −0.02 and 0.02, and read 0.136 and 0.039, the first the affine's read of the store alone and the second the collapse. So at half the width the level is the width-12 store's within the seed spread and the plateau the deletion-only graph had removed returns on a third of the seeds, as it did under the solve head (round 134): the backbone is not too big for this target at width 12, and width 6 is where the escape starts to fail. With the width-90 reads of rounds 122 to 132 the size axis is now read from 6 to 90 and the level does not move on it.

Round 145 is read (job 2799, 3 h 34 min): the size question from the regularization side, the store arm on the deletion-only graph with the transformer's dropout raised from the incumbent 0.1 to 0.3, paired against the store at 0.1. It reads 0.145 (sd 0.018) against 0.121 (+0.024 paired, sd 0.040, above on four of six), the highest single-seed rank of any arm on this target (the HiGHS store 0.130, the width-90 store 0.134), with a top-50 enrichment of 5.03 against 4.93, the highest single-seed call, and ensembled over six seeds 0.149 with a top-50 of 5.79 against 0.131 and 5.59, the highest width-12 ensemble rank by 0.001 over the HiGHS store. Five seeds select off the plateau at epochs 39 to 81 with validation Pearson 0.20 to 0.25 and read 0.135 to 0.180, the tightest escaped band of any arm; the sixth sits on the plateau to epoch 149 and reads 0.133, the affine alone. So the backbone that every width from 6 to 90 left at one level moves when it is regularized harder: the paired gain is inside the seed spread on the mean and on four seeds of six, but the level, the call and the ensemble are each the best a single width-12 arm has read, and the direction the user's question pointed at, a model too free for 3,494 genotypes, reads as measured. Rounds 154 and 155 (w171, w172) put dropout 0.3 on the HiGHS store under the release rule, the three best levers on one cell, and read dropout 0.5 on the store arm.

Round 147 is read (job 2801, 3 h 39 min): the probabilistic metabolite side in the affine slot, the mean concentration over six draws of each deletion's driving-force polytope (Section 3) with the point store's statistics and the count of constrained metabolites, on the deletion-only graph. It reads 0.125 (sd 0.041) against the affine-free control's 0.058 (+0.067 paired, sd 0.047, above on six of six), against the GLPK flux store's 0.121 (+0.004, sd 0.047, four of six) and against the driving-force point's 0.118 (+0.008, sd 0.054, three of six), top-50 3.69 against 3.39, 4.93 and 4.06, and ensembled over six seeds 0.151 with a top-50 of 5.59 against 0.082, 0.131 and 0.133, the highest six-seed rank of any width-12 arm. Every seed selects off the plateau, at epochs 10 to 76 on validation Pearson 0.15 to 0.28, and reads 0.055 to 0.168. So the region's mean carries what its point carries and what the flux vector carries, on every seed above the control, and averages to the best rank a width-12 cell has reached, while its single-seed call is the weakest of the three: the second law's region, read as its centroid, is a third representation at the flux store's level, with more to give the ensemble than the point and less to give the call. The width of the region, the spread store the ladder read at 0.138 by nearest neighbors, has not been read in the transformer.

Round 149 is read (job 2804, 3 h 35 min): round 131's release rule on round 135's store, the linear adjustment store with the affine frozen until epoch 5 and until the trailing validation Pearson reaches 0.15, on the deletion-only graph. It reads 0.106 (sd 0.034) against the adjustment store with the affine from epoch 0 at 0.107 (−0.001 paired, sd 0.038, above on three of six) and against the GLPK store under the same rule at 0.110 (−0.004, sd 0.063, two of six), top-50 4.53 against 4.99 and 4.66, and ensembled over six seeds 0.105 with a top-50 of 5.59 against 0.115 and 5.79 and 0.120 and 6.19. The release came at epochs 5, 5, 22, 30, 9 and 5, and the seeds select at 2, 51, 76, 43, 6 and 32, two of them before the release on early validation peaks of 0.26 and 0.17, reading 0.117 and 0.058. So the rule that kept the GLPK store's level with the best ensemble call (round 131) neither adds to nor takes from the adjustment store's level and costs its call a point of enrichment on single seeds and on the ensemble: the adjustment store, anchored to one wild-type vector and free of the vertex noise, did not need the backbone to escape first, and the release rule's gain in round 131 reads as specific to the degenerate store it was measured on.

Round 150 is read (job 2805, 3 h 39 min): the GO multi-hot in the affine slot, 2,362 propagated terms over every annotated gene (Section 3), on the deletion-only graph, the interface the flux stores enter through and the annotation the backbone never saw. It reads 0.202 (sd 0.001 over seeds) against the affine-free control's 0.058 (+0.144 paired, sd 0.075, above on six of six) and against the GLPK store's 0.121 (+0.082, sd 0.025, six of six), with a top-50 enrichment of 5.03 against 3.39 and 4.93, and ensembled over six seeds 0.202 with a top-50 of 4.99: the highest single-seed rank of any transformer arm on this target, at the level the GO perceptron read on its own (0.200), with the best single-seed call tied with round 145, and the first arm whose ensemble adds nothing, because the six seeds read the same number. Every seed selects at epochs 8 to 12 on validation Pearson 0.25, the validation loss bottoms at epochs 1 to 3 and rises by 0.13 to 0.18 by epoch 199, and the seeds differ in the third decimal: the affine's linear read of the binary columns is the whole cell, learned in ten epochs and memorized thereafter, and the backbone, which escapes to the same 0.25 on every seed, adds nothing the affine did not already have. So the transformer is not what limits the annotation family on this target; the annotation it was given was. What remains is whether the flux vector adds to GO in the one model,

round 151, and whether either adds to the other's call.

Round 151 is read (job 2806, 3 h 38 min): the HiGHS parsimonious deletion vector with the GO multi-hot appended, 4,177 flux columns (the wild-type row for a gene outside the GEM) and 2,362 annotation columns in one affine over every annotated gene, on the deletion-only graph. It reads 0.211 (sd 0.007) against the HiGHS store alone at 0.130 (+0.081 paired, sd 0.032, above on six of six) and against GO alone at 0.202 (+0.009, sd 0.007, five of six), with a top-50 enrichment of 4.83 against 4.46 and 5.03, and ensembled over six seeds 0.212 with a top-50 of 4.99 against 0.148 and 0.202, the highest rank of the study on single seeds and on the ensemble. Every seed selects at epochs 8 to 13 on validation Pearson 0.24 to 0.26 and reads 0.202 to 0.220, the round-150 pattern with a wider band. So the flux vector adds to the annotation in the one model, by 0.009 on five seeds of six against a paired spread of 0.007, where the perceptron on the same two blocks read no gain (0.198 against 0.200 with a spread of 0.041): the affine on the logit is the interface on which the two families add, and what the flux store contributes beside GO is a tenth of what GO contributes beside the flux store. The call does not follow the rank: 4.83 sits under GO alone's 5.03, as the flux store's call sat under the control's in every round that read it beside a stronger input.

Round 152 is read (job 2807, 3 h 40 min): round 131's release rule on the HiGHS parsimonious store, the affine frozen until epoch 5 and until the trailing validation Pearson reaches 0.15, on the deletion-only graph. It reads 0.108 (sd 0.021) against the HiGHS store with the affine from epoch 0 at 0.130 (−0.022 paired, sd 0.042, above on two of six) and against the GLPK store under the same rule at 0.110 (−0.002, sd 0.058, two of six), top-50 3.99 against 4.46 and 4.66, and ensembled over six seeds 0.139 with a top-50 of 4.99 against 0.148 and 5.79 and 0.120 and 6.19. The release came at epochs 5, 5, 22, 30, 9 and 5, the same epochs as round 149 on the same seeds, and the seeds select at 2, 2, 95, 121, 55 and 43, two of them before the release on early validation peaks of 0.26 and 0.21, reading 0.117 and 0.126. The HiGHS store's plateau seed, which read 0.160 from the affine alone at epoch 2 with the backbone at validation 0.02, releases at epoch 9 under the rule, selects at 55 and reads 0.117. So the rule takes from the clean vector what it added to the degenerate one: the HiGHS store under the rule reads at the GLPK store's level under the rule, below the HiGHS store free from epoch 0 on four seeds of six, on the call and on the ensemble, and the pair with round 149 closes the question the rule opened in round 131. The gain was the vertex noise's, held off until the backbone had something to say; a store without that noise is read best from the first epoch.

Round 154 is read (job 2809, 3 h 40 min): the three best levers on one cell, the HiGHS parsimonious store (round 139), the affine released on escape (round 131) and transformer dropout 0.3 (round 145), on the deletion-only graph. It reads 0.125 (sd 0.036) against the same cell without the dropout (round 152) at 0.108 (+0.017 paired, sd 0.018, above on five of six) and against the GLPK store at dropout 0.3 (round 145) at 0.145 (−0.020, sd 0.052, three of six), top-50 4.36 against 3.99 and 5.03, and ensembled over six seeds 0.148 with a top-50 of 4.39 against 0.139 and 4.99 and 0.149 and 5.79. Dropout delays the escape the rule waits for: the release came at epochs 17, 5, 24, 44, 18 and 5 against 5, 5, 22, 30, 9 and 5 without it, every seed selects after its release, at 54 to 136 on validation Pearson 0.21 to 0.25, and the seeds read 0.060 to 0.155, the lowest the one released last. So dropout gives back about half of what the release rule took from the HiGHS store (0.130 free from epoch 0, 0.108 under the rule, 0.125 under the rule with dropout), and the three levers together sit under the two that do not fight: the GLPK store at dropout 0.3 keeps the higher rank and the higher call, on single seeds and on the ensemble. The lever that costs on the clean store is the release, and dropout does not repair it.

Round 155 is read (job 2810, 3 h 38 min): the dropout curve's third point, the store arm on the deletion-only graph at transformer dropout 0.5, paired against 0.3 (round 145) and 0.1 (the incumbent). It reads 0.125 (sd 0.017) against 0.145 at 0.3 (−0.019 paired, sd 0.019, above on two of six) and against 0.121 at 0.1 (+0.005, sd 0.039, three of six), top-50 4.16 against 5.03 and 4.93, and ensembled over six seeds 0.127 with a top-50 of 5.59 against 0.149 and 5.79 and 0.131 and 5.59. Three seeds never leave the plateau: their validation Pearson sits at −0.03 to −0.02 through epoch 199, they select at 149 to 181 and read 0.124 to 0.133, the affine alone; the three that escape select at epochs 68 to 156 on validation Pearson 0.16 to 0.24 and read 0.094 to 0.140, no higher. So the curve turns between 0.3 and 0.5: at 0.5 the regularization that lifted the backbone at 0.3 keeps half the seeds from escaping at all, and the seeds that do escape read at the affine's level, so the arm sits at the incumbent's level with the incumbent's spread and a weaker call. Dropout 0.3 stands as the setting, and its gain in round 145 reads as a maximum of the curve, not a monotone trend to push further.

Round 156 is read (job 2812, 3 h 41 min): the width of the region, the driving-force polytope's mean concentration with the per-metabolite spread of $\ln c$ over draws appended (5,666 columns; the ladder read this store at 0.138 by nearest neighbors against the mean's 0.127), in the affine slot on the deletion-only graph. It reads 0.116 (sd 0.041) against the polytope mean alone (round 147) at 0.125 (−0.010 paired, sd 0.062, above on three of six) and against the GLPK store's 0.121 (−0.005, sd 0.048, four of six), top-50 3.73 against 3.69 and 4.93, and ensembled over six seeds 0.155 with a top-50 of 4.59 against 0.151 and 5.59 and 0.131 and 5.59, the highest six-seed rank of any flux-side width-12 arm (the polytope mean 0.151, dropout 0.3 0.149, the HiGHS store 0.148). Two seeds escape only partly, selecting at epochs 38 and 118 on validation Pearson 0.09 and 0.10 and reading 0.061 and 0.089; the other four select at 59 to 152 on 0.23 to 0.30 and read 0.097 to 0.164. So the spread adds nothing to the mean on single seeds, inside a spread twice the difference, and the seeds disagree more than the mean's did; averaged, that disagreement is what the ensemble gains

from, and the region's width joins the region's mean as a representation whose value is in the ensemble rather than in any one read.

Round 158 is read (job 2813, 3 h 40 min): the GO affine under round 131's release rule, frozen until epoch 5 and until the trailing validation Pearson reaches 0.15, on the deletion-only graph; round 150 read the affine memorizing the annotation in ten epochs before the backbone moved, and the release asks whether a backbone that escapes first adds anything to that read. It reads 0.199 (sd 0.007) against the GO affine from epoch 0 at 0.202 (−0.003 paired, sd 0.007, above on four of six) and against the GLPK store under the same rule at 0.110 (+0.089, sd 0.047, six of six), top-50 4.89 against 5.03 and 4.66, and ensembled over six seeds 0.206 with a top-50 of 5.19 against 0.202 and 4.99 and 0.120 and 6.19. The release came at epochs 5, 5, 20, 42, 8 and 27, and every seed selects 6 to 11 epochs after it, at 14 to 48, on validation Pearson 0.24 to 0.28: the affine learns the annotation in the same ten epochs whether the backbone has escaped or not, the two seeds released last read lowest (0.191 and 0.189), and the seeds now differ where round 150's did not (sd 0.007 against 0.001), so the ensemble gains 0.003 over the identical-seed ensemble and its call rises to 5.19. So the backbone escaping first adds nothing to the annotation's single read and takes a little from it when the escape is late; what the rule buys is seed diversity, worth 0.003 on the ensemble, on an arm whose value was never in the backbone.

Round 159 is read (job 2814, 3 h 40 min): the GO affine with transformer dropout 0.3, round 145's lever on the annotation arm, on the deletion-only graph. It reads 0.202 (sd 0.002) against the GO affine at dropout 0.1 at 0.202 (−0.001 paired, sd 0.001, above on no seed) and against the GLPK store at dropout 0.3 at 0.145 (+0.057, sd 0.019, six of six), top-50 4.89 against 5.03 and 5.03, and ensembled over six seeds 0.202 with a top-50 of 4.79 against 0.202 and 4.99 and 0.149 and 5.79. Every seed selects at epochs 8 to 14 on validation Pearson 0.24 to 0.25 and reads 0.198 to 0.203, round 150's pattern to the third decimal. So the dropout that lifted the store arm by 0.024 moves the annotation arm by less than a thousandth, the same way on every seed: the lever acts on the backbone, and on this arm the backbone contributes nothing for it to act on. Taken with round 158, the annotation arm is the affine and the affine alone, and no setting of the transformer around it has moved it.

Round 161 is read (job 2816, 3 h 40 min): the campaign's two best levers on one cell, the HiGHS vector with GO appended in the affine (round 151) with transformer dropout 0.3 (round 145), on the deletion-only graph. It reads 0.212 (sd 0.007) against the same affine at dropout 0.1 at 0.211 (+0.000 paired, sd 0.001, above on two of six) and against GO alone at dropout 0.3 at 0.202 (+0.010, sd 0.007, five of six), top-50 4.89 against 4.83 and 4.89, and ensembled over six seeds 0.213 with a top-50 of 4.99 against 0.212 and 4.99 and 0.202 and 4.79. Every seed selects at epochs 7 to 13 on validation Pearson 0.24 to 0.26 and reads 0.202 to 0.220. So the two levers do not add: dropout leaves the flux-and-annotation affine where it was, to the fourth decimal on the mean, as it left the annotation affine in round 159, and the flux vector's gain beside GO reproduces at dropout 0.3 (0.010 on five seeds of six, against 0.009 at 0.1). The best cell of the study is the affine over the two blocks, at either dropout, and what the transformer is set to around it does not enter the read.

Round 163 is read (job 2821, 3 h 39 min): the physics stack on the annotation, the HiGHS vector, GO and the driving-force polytope's mean concentration (round 147) in one affine over every annotated gene, 9,374 columns joined by `target_store_concat.py` with the polytope's wild-type row for a gene outside the GEM, on the deletion-only graph. It reads 0.198 (sd 0.011) against the same affine without the polytope (round 151) at 0.211 (−0.013 paired, sd 0.009, above on one of six) and against the polytope mean alone at 0.125 (+0.073, sd 0.045, six of six), top-50 4.46 against 4.83 and 3.69, and ensembled over six seeds 0.201 with a top-50 of 4.59 against 0.212 and 4.99 and 0.151 and 5.59. Every seed selects at epochs 4 to 15 on validation Pearson 0.25 to 0.27 and reads 0.180 to 0.213, a band five times wider than round 151's. So the 2,835 concentration columns take 0.013 from the affine that carried GO and the flux vector, on five seeds of six, and take its call with it, where the flux vector's 4,177 columns had added 0.009 beside GO: the metabolite side, which matched the flux vector in the affine on its own (round 147), does not add to the annotation the way the flux vector does, and the stack reads under GO alone (0.202). Hypothesis (untested): the affine's linear read over 3,494 training genotypes, most of them the wild-type row on every physics column, is diluted rather than informed by a third block whose rows the flux block already distinguishes.

Round 164 is read (job 2822, 3 h 36 min): six more seeds, 2706 to 2711, of round 151's cell, the HiGHS vector with GO appended in the affine, so the study's best arm has a twelve-seed level and ensemble. The second six read 0.218 (sd 0.009) against the first six's 0.211 (sd 0.007); over twelve the level is 0.214 (sd 0.008, standard error 0.002), the seeds select at epochs 6 to 13 and read 0.202 to 0.232, and the twelve-seed ensemble reads 0.216 with a top-50 of 4.99 against 0.212 and 4.99 for six, the single seeds' mean call 4.98. So the best arm's level holds on twelve seeds, at 0.214 with a standard error of 0.002, which puts it 0.011 above the random forest over the same columns (0.204, Section 3) and 0.019 above the nearest-neighbor read of the HiGHS store (0.195), the two closest backbone-free reads on this split; the ensemble gains 0.004 from the second six and the call does not move.

Round 166 is read (job 2824, 3 h 39 min): the same stack with the region's width, the HiGHS vector, GO and the polytope's mean with the per-metabolite spread of $\ln c$, 12,205 columns in one affine on the deletion-only graph. It reads 0.193 (sd 0.015) against the flux-and-annotation affine (round 151) at 0.211 (−0.018 paired, sd 0.016, above on two of six) and against the same stack with the mean alone (round 163) at 0.198 (−0.005, sd 0.018, two of six), top-50

4.23 against 4.83 and 4.46, and ensembled over six seeds 0.201 with a top-50 of 4.59, the same ensemble as the mean alone. The seeds select at epochs 9 to 27 on validation Pearson 0.22 to 0.28 and read 0.175 to 0.213, twice round 163's band and three times round 151's. So the second physics block costs what the first cost again, and the cost grows with the columns added: 2,835 concentration columns took 0.013, another 2,831 spread columns take 0.005 more, while the seed spread doubles at each step. The metabolite side does not enter this affine's read at any width, and the study's best cell remains the flux vector with the annotation and nothing else.

Round 167 is read (job 2825, 3 h 39 min): which flux program adds most beside the annotation, the linear adjustment store (round 135) in place of the parsimonious vector in the flux-and-annotation affine, 6,539 columns on the deletion-only graph. It reads 0.208 (sd 0.003) against GO alone at 0.202 (+0.006 paired, sd 0.003, above on six of six) and against the parsimonious vector with GO at 0.211 (−0.004, sd 0.006, two of six), top-50 4.46 against 5.03 and 4.83, and ensembled over six seeds 0.209 with a top-50 of 4.39 against 0.202 and 0.212. The seeds select at epochs 8 to 16 on validation Pearson 0.24 to 0.27 and read 0.205 to 0.212. So the adjustment store adds to the annotation as the parsimonious vector does, on every seed, and adds a little less: what the flux side contributes beside GO is 0.006 from the store anchored to one wild-type vector against 0.009 from the parsimonious rebuild, a difference inside the paired spread. Both physically stated flux programs enter the annotation's read where the metabolite programs do not, and the parsimonious vector stays the best of them.

Round 169 is read (job 2828, 3 h 40 min): the one setting left that acts on the affine's own read, its weight decay. Round 151's cell, the HiGHS vector with GO in the affine on the deletion-only graph, with the affine in its own optimizer group at weight decay 0.1 and the full rate (round 119 read decay and a tenth of the rate together on the degenerate store and lost the gain, so the rate is left alone). It reads 0.210 (sd 0.007) against no decay at 0.211 (−0.002 paired, sd 0.008, above on four of six), top-50 4.66 against 4.83, and ensembled over six seeds 0.211 with a top-50 of 4.79 against 0.212 and 4.99. The seeds select at epochs 8 to 16 on validation Pearson 0.24 to 0.27 and read 0.199 to 0.221; on five seeds the arm and its control select at the same epoch and differ by under 0.004, and the sixth selects seven epochs later and reads 0.018 under. So a thousandfold increase in the decay on the affine's weights leaves the selected epochs and the read where they were. The affine over the annotation and the flux vector is read within its first sixteen epochs, and regularizing it does not move that read either.

Round 170 is read (job 2829, 3 h 40 min): the second point of the curve, the same cell at affine weight decay 1.0. It reads 0.204 (sd 0.012) against no decay at 0.211 (−0.007 paired, sd 0.013, above on one of six) and against decay 0.1 at 0.210 (−0.005, sd 0.013, two of six), top-50 4.76 against 4.83 and 4.66, and ensembled over six seeds 0.208 with a top-50 of 4.99 against 0.212 and 0.211. The seeds select at epochs 34 to 114 on validation Pearson 0.25 to 0.27, against 8 to 16 for both lighter settings, and read 0.190 to 0.218, a spread of 0.012 against 0.007. So the decay curve runs 0.211, 0.210, 0.204 from none to 0.1 to 1.0: flat, then down, with the heavy setting slowing the affine so its read arrives fifty epochs later and spreads more across seeds. The affine's own regularization does not lift the annotation arm any more than the transformer's settings did, and the study's best cell stays round 151's as built.

Round 171 is read (job 2830, 3 h 36 min): the two flux programs that each add to the annotation, read together. The parsimonious vector and the adjustment store both appended to GO in one affine, 10,716 columns on the deletion-only graph. It reads 0.216 (sd 0.007) against the parsimonious vector with GO (round 151) at 0.211 (+0.004 paired, sd 0.003, above on five of six) and against the adjustment store with GO (round 167) at 0.208 (+0.008, sd 0.006, five of six), top-50 4.73 against 4.83 and 4.46, and ensembled over six seeds 0.217 with a top-50 of 4.59 against 0.212 and 0.209. The seeds select at epochs 9 to 14 on validation Pearson 0.25 to 0.26 and read 0.206 to 0.223. So the two programs add: what the adjustment store carries beside the annotation is not all inside what the parsimonious vector carries, and the two together give the highest single-seed rank of the study, 0.216, at a paired spread of 0.003 against the previous best. The call does not follow, 4.73 against 4.83 on single seeds and 4.59 against 4.99 ensembled. The flux side enters the annotation's read a second time where the metabolite side (rounds 163, 166) did not enter it once.

Round 172 is read (job 2831, 3 h 40 min): six more seeds, 2706 to 2711, of the annotation-only arm, GO alone in the affine (round 150), so it has the twelve-seed level the flux-and-annotation arm has had since round 164. The second six read 0.204 (sd 0.002) against the first six's 0.202 (sd 0.001); over twelve, 0.203 (sd 0.002, standard error 0.0005), the seeds reading 0.200 to 0.207 and selecting at epochs 7 to 14; the twelve-seed ensemble reads 0.203 with a top-50 of 4.99, as the six-seed one did. The flux vector's gain beside GO, read paired over the same twelve seeds, is +0.012 (sd 0.009), above on eleven of twelve, where the first six gave +0.009 on five of six: 0.214 against 0.203. The annotation arm is the study's most reproducible cell, its twelve seeds inside 0.007 of one another, and what the flux vector adds to it is four times its seed spread.

Round 173 is read (job 2832, 3 h 41 min): six more seeds of the affine-free control on the deletion-only graph (round 132's cell), so the annotation arm's gain is read paired over twelve seeds, the comparison the amino-acid study made in its round 168. The control's second six read 0.094 (sd 0.017) against the first six's 0.058 (sd 0.075); over twelve 0.076 (sd 0.055), the seeds reading −0.089 to 0.126 and selecting at epochs 3 to 69, since a control seed's score depends on whether and when it escapes the plateau. The GO affine against it, paired over the same twelve seeds, is +0.127 (sd 0.054), above on twelve of twelve, 0.203 against 0.076, top-50 4.99 against 4.06. The twelve-seed ensemble of the control reads 0.102 with a top-50 of 5.59, against 0.082 and 4.79 from its first six: averaging the control's seed noise away is

worth 0.026 of rank and 0.8 of call, and the annotation's twelve-seed ensemble, 0.203, stays twice the control's. On this target the amino-acid study's finding does not repeat: there, the control's ensemble passed the annotation's because the annotation's gain (0.009) was smaller than the control's seed noise; here the gain is 0.127 and five times what the control's ensemble recovers.

Rounds 175 and 176 are read (jobs 2836 and 2837, 3 h 38 min and 3 h 42 min): the measured side in the transformer's affine, the strain's own Mülleder amino-acid pool (Section 3), appended to GO and to GO with the HiGHS vector. With GO the pool reads 0.208 (sd 0.002) against GO alone at 0.202 (+0.006 paired, sd 0.002, above on six of six), top-50 4.89 against 5.03, ensembled 0.209 with a top-50 of 4.79; the seeds select at epochs 7 to 13. Beside the HiGHS vector the pool reads 0.208 (sd 0.009) against 0.211 without it (−0.003, sd 0.014, two of six) and 0.000 against the pool with GO alone (three of six), with the seed spread quadrupled; ensembled 0.209 and 4.79. So the affine reads the nineteen columns as the ridge does, a small addition to the annotation, where the ladder's nearest neighbors and forest read them at 0.285 and 0.280 on their own: the pool's signal is there and it is not linear in the concentrations. The transformer's read of the pool is bounded by the slot it enters through, not by the data; the pool alone in the affine and through a hidden layer of 64 are queued to say which (179, 180).

Round 177 is read (job 2839, 3 h 40 min): six more seeds of round 171's cell, both flux programs beside GO, so the study's best single-seed arm has the twelve-seed level its two comparators have. The second six read 0.219 (sd 0.003) against the first six's 0.216 (sd 0.007); over twelve, 0.217 (sd 0.005, standard error 0.0015), the seeds 0.206 to 0.223 selecting at epochs 5 to 14; the twelve-seed ensemble reads 0.218 with a top-50 of 4.79 against 0.217 and 4.59 from six. Paired over the same twelve seeds against the parsimonious vector with GO, the second program's gain is +0.003 (sd 0.007), above on eight of twelve, where the first six gave +0.004 on five of six at a paired spread of 0.003: on the second six the two arms read the same, 0.219 against 0.218. So the gain of round 171 does not hold at the size the first six gave it; the adjustment store beside the parsimonious vector is worth 0.003 within a paired spread of 0.007, and the two arms share one level, 0.214 to 0.217 on twelve seeds.

Round 178 is read (job 2840, 3 h 40 min): the three families in one affine, GO with both flux programs and the strain's measured amino-acid pool, 10,735 columns on the deletion-only graph. It reads 0.218 (sd 0.002) against both flux programs with GO (round 171) at 0.216 (+0.002 paired, sd 0.008, three of six) and against the pool with GO and the parsimonious vector (round 176) at 0.208 (+0.010, sd 0.009, five of six), top-50 4.63 against 4.73 and 4.66, ensembled 0.218 with a top-50 of 4.59. The seeds select at epochs 10 to 14 on validation Pearson 0.25 to 0.27 and read 0.216 to 0.221, the tightest band of any store arm. So the pool adds nothing beside the two flux programs, as it added nothing beside one (round 176), and every linear combination of the annotation with the flux side and the measured side reads between 0.208 and 0.218: the affine's ceiling on this input family, with the measured side's nonlinear signal (0.285 by nearest neighbors on the pool alone) the one thing it does not reach.

Round 179 is read (job 2841, 3 h 36 min): the strain's measured amino-acid pool alone in the affine, nineteen columns and no annotation, on the deletion-only graph. It reads 0.195 (sd 0.029) against the affine-free control's 0.058 (+0.137 paired, sd 0.088, six of six) and against GO alone at 0.202 (−0.008, sd 0.030, four of six), top-50 2.53 against 3.39 and 5.03, the lowest call of any store arm, ensembled 0.211 with a top-50 of 3.00. The seeds select at epochs 13 to 32 on validation Pearson 0.33 to 0.41, the highest validation correlation of any arm in the study against the annotation arm's 0.26, and read 0.150 to 0.220, a spread twenty times the annotation arm's. Beside the ladder on the same columns the affine sits above the ridge (0.149) and 0.09 under the nearest neighbors (0.285) and the forest (0.280): the pool's linear read in the transformer is a little more than a ridge and far short of what the distance models take from the same nineteen numbers, and its validation-to-test gap is the widest of any arm. So the affine's ceiling of round 178 is the slot's, not the backbone's, and the hidden layer of 64 over the same columns (round 180) is the direct test of whether the transformer can read the pool nonlinearly at all.

Round 180 is read (job 2842, 3 h 41 min): the pool alone through one GELU hidden layer of 64 in the affine slot, the same nineteen columns as round 179 read nonlinearly. It reads 0.240 (sd 0.048) against the linear affine's 0.195 (+0.045 paired, sd 0.047, five of six) and against the control's 0.058 (+0.182, sd 0.099, six of six), the highest single-seed mean of any transformer arm in the study, above every linear cell's 0.208 to 0.218; top-50 4.03 against 2.53 and 3.39. The six-seed ensemble reads 0.292 with a top-50 of 5.79, above the ladder's nearest neighbors (0.285) and forest (0.280) on the same columns and 0.074 above the best linear ensemble (0.218, round 178). The seeds read 0.159 to 0.300 on validation Pearson 0.35 to 0.38, and select at epochs 13 to 197: one seed selects three epochs before the budget's end, so its peak is not established and the level of this arm is a lower bound on what a longer budget reads. The hidden layer does what round 134's hidden layer of 256 over the flux store did not: here the nonlinear head lifts the read by 0.045 where there it lost 0.02, so the pool carries signal that a hidden layer recovers and the flux vector does not. The annotation with the pool through the same hidden layer, the pool through a layer of 256, the three families through 64 and six more seeds of this arm are queued (rounds 185 to 188).

Round 181 is read (job 2843, 3 h 40 min): the minimum-norm flux store alone in the affine, the ℓ_2 point at maximal growth (Section 3) with no vertex noise. It reads 0.093 (sd 0.013) against the HiGHS parsimonious store on the same seeds (round 139) at 0.130 (−0.037 paired, sd 0.043, one of six) and against the control's 0.058 (+0.035, sd 0.076, four of six), top-50 4.59 against 4.46 and 3.39, ensembled 0.094 with a top-50 of 5.19. The seeds read 0.070 to 0.107 and

select at epochs 3 to 64. On the ladder the same store reads 0.068, 0.126 and 0.133 by ridge, nearest neighbors and forest against the HiGHS store's 0.098, 0.195 and 0.117, so the transformer's affine and the distance models agree: the unique interior point carries less rank than the sparse vertex, and the vertex noise the release rule was built around is not what the affine was reading.

Round 182 is read (job 2844, 3 h 40 min): the minimum-norm store appended to GO, 6,539 columns. It reads 0.204 (sd 0.005) against the HiGHS vector with GO (round 151) at 0.211 (−0.007 paired, sd 0.007, zero of six) and against GO alone at 0.202 (+0.002, sd 0.005, four of six), top-50 4.36 against 4.83 and 5.03, ensembled 0.206 with a top-50 of 4.39. The seeds read 0.198 to 0.212 and select at epochs 8 to 16, the annotation arm's band. So the flux side's gain beside the annotation, +0.009 on six seeds and +0.012 on twelve for the ℓ_1 vector, is absent for the ℓ_2 point on every seed: whatever the affine reads from the flux vector beside GO lives in the sparse solution, and the nearest-neighbor read of the same columns at 0.208, the best kNN of any genotype-side representation, does not reach the transformer.

Round 185 is read (job 2847, 3 h 33 min): GO with the measured pool, 2,381 columns, through the same hidden layer of 64. It reads 0.206 (sd 0.006) against the pool alone through the layer at 0.240 (−0.034 paired, sd 0.052, one of six), against the same columns in the linear affine at 0.208 (−0.002, sd 0.008, three of six) and against GO alone at 0.202 (+0.004, sd 0.007, three of six), top-50 4.59 against 4.03, 4.89 and 5.03, ensembled 0.208 with a top-50 of 4.99. The seeds select at epochs 2 to 4 on validation Pearson 0.22 to 0.29, the earliest selection of any arm, where the pool alone through the layer selects at 13 to 197 on 0.35 to 0.38. So the annotation appended to the pool takes the nonlinear read back to the annotation's own level: with GO in the layer the arm behaves as the GO affine does, in its level, its seed band and its early selection, and the pool's 0.034 is gone. The ladder read the same thing on the same columns, ridge and nearest neighbors falling when the 2,362 GO columns joined the nineteen and only the forest keeping the pool's rank (Section 3). A plausible reason, unverified, is that a layer of 64 over 2,381 columns fits the annotation's columns in the first epochs and the nineteen pool columns never set its selection; a layer that sees the pool and the annotation through separate slots is the arm that would separate the two.

Round 186 is read (job 2848, 3 h 34 min): six more seeds of the pool through the hidden layer of 64. The second six read 0.268 (sd 0.032) against the first six's 0.240, and over twelve the arm reads 0.254 (sd 0.041, standard error 0.012), seeds 0.159 to 0.300, selecting at epochs 4 to 197; the twelve-seed ensemble reads 0.297 with a top-50 of 5.79. Paired over the same twelve seeds the arm is +0.178 over the affine-free control on twelve of twelve and +0.051 over GO alone on eleven of twelve, with a top-50 of 4.24 against the annotation's 4.99. So the hidden-layer read of the pool holds at its second size: the highest single-seed level and the highest ensemble of any transformer arm, the ensemble above the ladder's nearest neighbors and forest on the same columns, the spread twenty times the annotation arm's, and the call under the annotation's on every count.

Round 187 is read (job 2853, 3 h 30 min): the pool alone through a hidden layer of 256. It reads 0.279 (sd 0.023) against the layer of 64 at 0.240 (+0.039 paired, sd 0.051, five of six) and against the linear slot at 0.195 (+0.084, sd 0.035, six of six), top-50 4.69 against 4.03 and 2.53, ensembled 0.288 with a top-50 of 5.19. The seeds read 0.242 to 0.303 and select at epochs 29 to 138 on validation Pearson 0.36 to 0.40, half the spread of the layer of 64 and the highest validation correlation of any arm. So the width of the nonlinear read matters on these nineteen columns: 64 to 256 adds 0.039 with the seed band halved, and the single-seed level now sits at the ladder's nearest-neighbor read of the same columns (0.285) with the ensemble above it. The layer of 128 and the layer of 512 are queued (rounds 194, 192), with six more seeds of this arm (193).

Round 188 is read (job 2854, 3 h 34 min): the three families, GO with both flux programs and the pool, through the hidden layer of 64. It reads 0.217 (sd 0.009) against the same columns in the linear affine at 0.218 (−0.002 paired, sd 0.009, two of six), against GO with the pool through the layer at 0.206 (+0.011, sd 0.014, five of six) and against the pool alone through the layer at 0.240 (−0.023, sd 0.046, one of six), top-50 4.53, ensembled 0.227 with a top-50 of 4.19. The seeds read 0.200 to 0.226 and select at epochs 5 to 20 on validation Pearson 0.25 to 0.27. With the annotation in the layer the read is again the annotation's, as in round 185, and the flux programs add their usual 0.01 beside it: the layer does not separate the nineteen pool columns from the 10,716 others.

Round 191 is read (job 2865, 3 h 33 min): the pool with the HiGHS parsimonious vector appended on the pool's own 4,678 genes, 4,196 columns with 815 genes carrying a flux row and the rest the flux wild type, through the hidden layer of 64. It reads 0.186 (sd 0.035) against the pool alone through the layer at 0.240 (−0.054 paired, sd 0.058, one of six) and against the HiGHS store alone in the linear affine at 0.130 (+0.056, sd 0.024, six of six), top-50 2.90 against 4.03 and 4.46, ensembled 0.207 with a top-50 of 4.59. The seeds read 0.138 to 0.234 and select at epochs 29 to 66 on validation Pearson 0.34 to 0.37. So the flux columns take from the pool's read as the annotation columns did, by 0.054 against the annotation's 0.034, and unlike the annotation they do not bring their own level with them: the layer over the joined store reads under both the pool through the layer and the annotation affine. Every store appended to the pool inside one hidden layer has cost it (rounds 185, 188, 191), so the pool's nonlinear read is best taken alone, and a head that reads the pool and the other families through separate slots is the arm that would test whether they add.

Round 190 is read (job 2864, 6 h 55 min): the pool through the hidden layer of 64 with a budget of 400 epochs, the same six seeds as round 180. Every seed selects the epoch it selected under 200 and reads the same value to the fourth

digit, 0.240 (sd 0.048) with a paired difference of exactly zero: the validation curve never passes its first 200 epochs' peak in the second 200, on the seed that selected at 197 as on the others. So round 180's level was not a lower bound set by the budget, and the selection at 197 was a peak, not a truncation.

Round 192 is read (job 2868, 3 h 35 min): the pool through a hidden layer of 512. It reads 0.243 (sd 0.048) against the layer of 256 at 0.279 (−0.036 paired, sd 0.059, one of six) and against the layer of 64 at 0.240 (+0.003, sd 0.072, three of six), top-50 4.36, ensembled 0.278 with a top-50 of 5.59; the seeds read 0.181 to 0.289 and select at epochs 2 to 91 on validation Pearson 0.36 to 0.39. The width past 256 gives back what 256 gained and brings the layer of 64's seed spread with it.

Round 193 is read (job 2869, 3 h 34 min): six more seeds of the pool through 256. The second six read 0.277 (sd 0.029) against the first six's 0.279, and over twelve the arm reads 0.278 (sd 0.025, standard error 0.007), seeds 0.241 to 0.313, selecting at epochs 29 to 141; the twelve-seed ensemble reads 0.296 with a top-50 of 5.19. Paired over the same twelve seeds the arm is +0.024 over the layer of 64 on nine of twelve, +0.075 over GO alone on twelve of twelve and +0.202 over the affine-free control on twelve of twelve, with a top-50 of 4.61 against the annotation's 4.99. The level holds at its second size with a band half the layer of 64's.

Round 194 is read (job 2870, 3 h 34 min): the pool through a hidden layer of 128. It reads 0.262 (sd 0.028) against the layer of 64 at 0.240 (+0.022 paired, sd 0.069, four of six) and against 256 at 0.279 (−0.017, sd 0.042, one of six), top-50 4.53, ensembled 0.279 with a top-50 of 5.19; the seeds read 0.214 to 0.290 and select at epochs 21 to 51 on validation Pearson 0.36 to 0.41. So the width curve of the pool's hidden read is 0.240, 0.262, 0.279, 0.243 at 64, 128, 256 and 512 on the same six seeds: a maximum at 256, which twelve seeds hold at 0.278, and the ensembles 0.292, 0.279, 0.288 and 0.278 within 0.015 of one another at every width, so the width sets the single-seed level and its spread more than what six seeds averaged read.

Rounds 197 and 198 are read (jobs 2877 and 2878, 3 h 36 min each): the two-slot head in the transformer's affine, GO's 2,362 columns through a linear map beside the pool's nineteen through a GELU hidden layer of 64 (197) or 256 (198), the two summed. At 64 it reads 0.224 (sd 0.003) against the same columns through one shared layer at 0.206 (+0.018 paired, sd 0.007, six of six), against the same columns in the linear affine at 0.208 (+0.016, sd 0.003, six of six), against GO alone at 0.202 (+0.022, sd 0.003, six of six) and against the pool alone through 64 at 0.240 (−0.016, sd 0.049, two of six), with a top-50 of 5.66, the best single-seed call of any transformer arm, ensembled 0.225 with a top-50 of 5.79. At 256 it reads 0.225 (sd 0.005), +0.001 against the two slots at 64 and −0.054 against the pool alone through 256 (0.279, sd 0.020, zero of six), ensembled 0.226. Both select at epochs 5 to 15 on validation Pearson 0.27 to 0.29, the annotation affine's schedule (8 to 12 on 0.25), where the pool alone through the layer selects at 29 to 141 on 0.36 to 0.40. So in the transformer the two slots do not do what they did in the perceptron: the head is the best arm with the annotation in it, above every GO cell on every seed with the tightest band and the best call, and it does not reach the pool alone, because the linear slot sets the selection before the pool's slot has trained. The perceptron trained full batch and selected at epochs 64 to 165; here the annotation's map at the backbone's rate peaks and turns within fifteen epochs. The linear slot at a tenth and a hundredth of the rate, in its own parameter group, is queued (rounds 201, 202).

Round 199 is read (job 2879, 3 h 34 min): the three families with two slots, GO and both flux programs (10,716 columns) through the linear map beside the pool's nineteen through a hidden layer of 256. It reads 0.239 (sd 0.003) against the same columns in the linear affine at 0.218 (+0.021 paired, sd 0.004, six of six), against the same columns through one shared layer of 64 at 0.217 (+0.023, sd 0.010, six of six), against GO alone in the linear slot beside the pool through 256 at 0.225 (+0.014, sd 0.004, six of six) and against the pool alone through 256 at 0.279 (−0.040, sd 0.022, zero of six), top-50 5.32, ensembled 0.241 with a top-50 of 5.19; the seeds read 0.235 to 0.243 and select at epochs 7 to 12 on validation Pearson 0.27 to 0.28. So the flux programs in the linear slot add 0.014 to the annotation's slot on every seed, as they added 0.012 to the annotation alone (round 172), and the two-slot head over the three families is the best arm of the study with the annotation in it, at 0.239 with the tightest band of any store arm; it selects on the same early schedule as the other two-slot heads and stays 0.04 under the pool alone.

Round 196 is read (job 2873, 7 h 30 min): the pool through 256 with a budget of 400 epochs, the same six seeds as round 187. As at width 64 (round 190), every seed selects the epoch it selected under 200 and reads the same value, 0.279 (sd 0.023), a paired difference of exactly zero on six of six: the validation curve never passes its first 200 epochs' peak. The width curve's maximum is a level, not a budget.

Round 204 is read (job 2886, 4 h 0 min): the pool through 256 read standardized instead of $\log(1+v)$, the perceptron sweep's encoding. It reads 0.182 (sd 0.017) against the $\log(1+v)$ read at 0.279 (−0.097 paired, sd 0.023, zero of six), top-50 3.99 against 4.69, ensembled 0.187; the seeds read 0.160 to 0.207 and select at epochs 2 to 8 on validation Pearson 0.22 to 0.27, where the $\log(1+v)$ cells select at 29 to 138 on 0.36 to 0.40. So the encoding that lifted the perceptron over the same columns (0.236 standardized against 0.173 raw, Section 3) costs the transformer's slot 0.10, and every seed selects within eight epochs, the schedule of the annotation's affine: in this trainer the pool is read best as the flux stores are read, unscaled.

Round 203 is read (job 2885, 3 h 58 min): the pool through 256 with dropout 0.3 inside the transformer, against 0.1. It

reads 0.279 (sd 0.018) against 0.279 (sd 0.023), a paired difference of −0.000 (sd 0.028, three of six), top-50 4.56 against 4.69, ensembled 0.293 with a top-50 of 5.59; the seeds select at epochs 32 to 138. On two seeds of six the two cells' pinned predictions agree to a Spearman of 1.0000 with the same selected epoch, although their training curves differ from the first epoch; on the other four they agree at 0.91 to 0.98, above the 0.91 that two seeds of the same arm agree to (`target_prediction_agreement.py`). A change inside the transformer that leaves a seed's prediction untouched did not reach the prediction, and Section 3.1 measures why: on those two seeds the backbone's head is a constant on the pinned genes and the prediction is the slot alone.

Round 200 is read (job 2880, 4 h 0 min): the two-slot head at a hidden width of 128, GO linear beside the pool through 128. It reads 0.225 (sd 0.005) against the pool alone through 128 at 0.262 (−0.037 paired, sd 0.026, one of six), against the two slots at 64 at 0.224 (+0.001, sd 0.005, three of six) and against the two slots at 256 at 0.225 (+0.000, sd 0.004, four of six), top-50 5.49 against 4.53, 5.66 and 4.99, ensembled 0.225 with a top-50 of 5.79; the seeds read 0.216 to 0.231 and select at epochs 7 to 10 on validation Pearson 0.27 to 0.29. So the width of the pool's slot does not enter the two-slot read at all, 0.224, 0.225 and 0.225 at 64, 128 and 256, where the pool alone moves from 0.240 to 0.262 to 0.279 over the same widths: with the annotation in the linear slot the cell selects on the annotation's schedule, before the pool's slot and the backbone have trained (Section 3.1), and the hidden slot's width is never read. The linear slot at a lower rate (201, 202) is the arm that can change this.

Table 1: Every arm paired by seed against its control on the same split (Merzbacher's 639 genes pinned to test, the 027 protocol). The statistic is the pinned-test Spearman at the epoch selected on validation Pearson (centered five-epoch moving average). Columns: seeds paired, the arm and control means with their standard deviations over seeds, the mean paired difference with its standard deviation, the seeds on which the arm scored higher, and the top-50 enrichment of high producers (arm / control). Written by `target_notes_tex_tables.py` from `results/transfer_readout.json` (`target_transfer_readout.py`).

arm against control	seeds	arm (sd)	control (sd)	difference (sd)	higher	top-50
whole-vector store affine against the control	6	0.1210 (0.0111)	0.0785 (0.0561)	+0.0425 (0.0489)	5/6	3.49 / 4.06
two summary columns against the control	6	0.1012 (0.0613)	0.0785 (0.0561)	+0.0226 (0.0723)	4/6	3.03 / 4.06
whole vector, affine decayed and slowed, against the control	6	0.0731 (0.0166)	0.0785 (0.0561)	−0.0054 (0.0575)	3/6	3.59 / 4.06
the growth block alone against the whole vector	6	0.0329 (0.0208)	0.1210 (0.0111)	−0.0881 (0.0291)	0/6	3.06 / 3.49
the production block alone against the whole vector	6	0.0911 (0.0396)	0.1210 (0.0111)	−0.0299 (0.0342)	1/6	3.63 / 3.49
the growth block alone against the control	6	0.0329 (0.0208)	0.0785 (0.0561)	−0.0456 (0.0635)	1/6	3.06 / 4.06
the production block alone against the control	6	0.0911 (0.0396)	0.0785 (0.0561)	+0.0126 (0.0210)	5/6	3.63 / 4.06
deletion-only cell graphs against the control	6	0.0582 (0.0746)	0.0785 (0.0561)	−0.0204 (0.0757)	2/6	3.39 / 4.06
six rule-count columns against the control	6	0.0743 (0.0594)	0.0785 (0.0561)	−0.0042 (0.0916)	2/6	2.96 / 4.06
width 90, two blocks, against the width-12 control	6	0.1087 (0.0533)	0.0785 (0.0561)	+0.0302 (0.0790)	4/6	4.73 / 4.06
width 90 with the store affine against the width-90 control	6	0.1339 (0.0201)	0.1087 (0.0533)	+0.0252 (0.0513)	4/6	4.09 / 4.73
width 90 with the store affine on deletion-only graphs against the width-90 control	6	0.1199 (0.0242)	0.1087 (0.0533)	+0.0112 (0.0670)	3/6	4.43 / 4.73
width 90 with the store affine on deletion-only graphs against the width-12 one	6	0.1199 (0.0242)	0.1208 (0.0249)	−0.0009 (0.0347)	3/6	4.43 / 4.93
width 90 on deletion-only graphs against the width-90 control	6	0.0921 (0.0425)	0.1087 (0.0533)	−0.0166 (0.0801)	2/6	4.49 / 4.73
width 90 with the store affine on deletion-only graphs against width 90 on deletion-only graphs	6	0.1199 (0.0242)	0.0921 (0.0425)	+0.0278 (0.0368)	4/6	4.43 / 4.49
deletion-only store, affine frozen to epoch 40, against the deletion-only store	6	0.0742 (0.0459)	0.1208 (0.0249)	−0.0466 (0.0534)	2/6	3.86 / 4.93
deletion-only store, affine frozen to epoch 40, against deletion-only graphs	6	0.0742 (0.0459)	0.0582 (0.0746)	+0.0160 (0.0679)	4/6	3.86 / 3.39
deletion-only store, affine from escape, against the deletion-only store	6	0.1104 (0.0496)	0.1208 (0.0249)	−0.0104 (0.0520)	1/6	4.66 / 4.93

Table 2: Table 1, continued (part 2 of 7).

arm against control	seeds	arm (sd)	control (sd)	difference (sd)	higher	top-50
deletion-only store, affine from escape, against deletion-only graphs	6	0.1104 (0.0496)	0.0582 (0.0746)	+0.0523 (0.0354)	6/6	4.66 / 3.39
deletion-only graphs with the store affine against deletion-only graphs	6	0.1208 (0.0249)	0.0582 (0.0746)	+0.0626 (0.0652)	6/6	4.93 / 3.39
deletion-only graphs with the store affine against the store affine	6	0.1208 (0.0249)	0.1210 (0.0111)	−0.0002 (0.0174)	4/6	4.93 / 3.49
vector with rule counts on deletion-only graphs against deletion-only graphs	6	0.1141 (0.0191)	0.0582 (0.0746)	+0.0559 (0.0561)	5/6	3.96 / 3.39
vector with rule counts on deletion-only graphs against the store affine there	6	0.1141 (0.0191)	0.1208 (0.0249)	−0.0067 (0.0228)	1/6	3.96 / 4.93
solve head (hidden 256) on deletion-only graphs against deletion-only graphs	6	0.0697 (0.0487)	0.0582 (0.0746)	+0.0115 (0.0737)	3/6	3.89 / 3.39
solve head (hidden 256) on deletion-only graphs against the store affine there	6	0.0697 (0.0487)	0.1208 (0.0249)	−0.0511 (0.0608)	1/6	3.89 / 4.93
linear MOMA store on deletion-only graphs against deletion-only graphs	6	0.1071 (0.0179)	0.0582 (0.0746)	+0.0489 (0.0637)	5/6	4.99 / 3.39
linear MOMA store on deletion-only graphs against the parsimonious store there	6	0.1071 (0.0179)	0.1208 (0.0249)	−0.0137 (0.0290)	1/6	4.99 / 4.93
HiGHS parsimonious store on deletion-only graphs against deletion-only graphs	6	0.1302 (0.0335)	0.0582 (0.0746)	+0.0720 (0.0673)	5/6	4.46 / 3.39
HiGHS parsimonious store on deletion-only graphs against the GLPK store there	6	0.1302 (0.0335)	0.1208 (0.0249)	+0.0094 (0.0413)	5/6	4.46 / 4.93
driving-force concentration store on deletion-only graphs against deletion-only graphs	6	0.1175 (0.0553)	0.0582 (0.0746)	+0.0593 (0.0554)	5/6	4.06 / 3.39
driving-force concentration store on deletion-only graphs against the GLPK store there	6	0.1175 (0.0553)	0.1208 (0.0249)	−0.0033 (0.0371)	2/6	4.06 / 4.93
sampled-cone mean store on deletion-only graphs against deletion-only graphs	6	0.0876 (0.0217)	0.0582 (0.0746)	+0.0295 (0.0696)	4/6	2.90 / 3.39
sampled-cone mean store on deletion-only graphs against the GLPK store there	6	0.0876 (0.0217)	0.1208 (0.0249)	−0.0332 (0.0241)	0/6	2.90 / 4.93
deletion-only store at width 6 against width 12	6	0.1131 (0.0377)	0.1208 (0.0249)	−0.0077 (0.0546)	4/6	4.19 / 4.93
deletion-only store at dropout 0.3 against 0.1	6	0.1447 (0.0176)	0.1208 (0.0249)	+0.0239 (0.0395)	4/6	5.03 / 4.93
GO multi-hot in the affine slot on deletion-only graphs against deletion-only graphs	6	0.2023 (0.0014)	0.0582 (0.0746)	+0.1441 (0.0745)	6/6	5.03 / 3.39

Table 3: Table 1, continued (part 3 of 7).

arm against control	seeds	arm (sd)	control (sd)	difference (sd)	higher	top-50
GO multi-hot in the affine slot on deletion-only graphs against the GLPK store there	6	0.2023 (0.0014)	0.1208 (0.0249)	+0.0815 (0.0245)	6/6	5.03 / 4.93
HiGHS store with GO appended against the HiGHS store alone	6	0.2114 (0.0068)	0.1302 (0.0335)	+0.0812 (0.0321)	6/6	4.83 / 4.46
HiGHS store with GO appended against GO alone	6	0.2114 (0.0068)	0.2023 (0.0014)	+0.0091 (0.0071)	5/6	4.83 / 5.03
HiGHS store with the affine released on escape against the HiGHS store	6	0.1082 (0.0210)	0.1302 (0.0335)	−0.0220 (0.0415)	2/6	3.99 / 4.46
HiGHS store with the affine released on escape against the GLPK store under the rule	6	0.1082 (0.0210)	0.1104 (0.0496)	−0.0023 (0.0575)	2/6	3.99 / 4.66
HiGHS store, release rule and dropout 0.3 against the same without the dropout	6	0.1248 (0.0361)	0.1082 (0.0210)	+0.0166 (0.0175)	5/6	4.36 / 3.99
HiGHS store, release rule and dropout 0.3 against the GLPK store at dropout 0.3	6	0.1248 (0.0361)	0.1447 (0.0176)	−0.0199 (0.0515)	3/6	4.36 / 5.03
deletion-only store at dropout 0.5 against 0.3	6	0.1253 (0.0165)	0.1447 (0.0176)	−0.0194 (0.0185)	2/6	4.16 / 5.03
deletion-only store at dropout 0.5 against 0.1	6	0.1253 (0.0165)	0.1208 (0.0249)	+0.0045 (0.0389)	3/6	4.16 / 4.93
MOMA store with the affine released on escape against the MOMA store	6	0.1062 (0.0339)	0.1071 (0.0179)	−0.0009 (0.0377)	3/6	4.53 / 4.99
MOMA store with the affine released on escape against the GLPK store under the rule	6	0.1062 (0.0339)	0.1104 (0.0496)	−0.0043 (0.0631)	2/6	4.53 / 4.66
driving-force polytope mean on deletion-only graphs against deletion-only graphs	6	0.1252 (0.0414)	0.0582 (0.0746)	+0.0671 (0.0465)	6/6	3.69 / 3.39
driving-force polytope mean on deletion-only graphs against the GLPK store there	6	0.1252 (0.0414)	0.1208 (0.0249)	+0.0044 (0.0469)	4/6	3.69 / 4.93
driving-force polytope mean on deletion-only graphs against the driving-force point	6	0.1252 (0.0414)	0.1175 (0.0553)	+0.0077 (0.0539)	3/6	3.69 / 4.06
driving-force polytope mean with the spread against the mean alone	6	0.1156 (0.0408)	0.1252 (0.0414)	−0.0096 (0.0623)	3/6	3.73 / 3.69
driving-force polytope mean with the spread against the GLPK store	6	0.1156 (0.0408)	0.1208 (0.0249)	−0.0052 (0.0482)	4/6	3.73 / 4.93
GO affine released on escape against the GO affine	6	0.1990 (0.0073)	0.2023 (0.0014)	−0.0033 (0.0069)	4/6	4.89 / 5.03
GO affine released on escape against the GLPK store under the rule	6	0.1990 (0.0073)	0.1104 (0.0496)	+0.0885 (0.0470)	6/6	4.89 / 4.66

Table 4: Table 1, continued (part 4 of 7).

arm against control	seeds	arm (sd)	control (sd)	difference (sd)	higher	top-50
GO affine at dropout 0.3 against the GO affine	6	0.2016 (0.0020)	0.2023 (0.0014)	−0.0006 (0.0007)	0/6	4.89 / 5.03
GO affine at dropout 0.3 against the GLPK store at dropout 0.3	6	0.2016 (0.0020)	0.1447 (0.0176)	+0.0569 (0.0187)	6/6	4.89 / 5.03
HiGHS store with GO appended at dropout 0.3 against 0.1	6	0.2115 (0.0068)	0.2114 (0.0068)	+0.0001 (0.0009)	2/6	4.89 / 4.83
HiGHS store with GO appended at dropout 0.3 against GO alone at 0.3	6	0.2115 (0.0068)	0.2016 (0.0020)	+0.0099 (0.0074)	5/6	4.89 / 4.89
HiGHS store with GO and the polytope mean against HiGHS store with GO	6	0.1981 (0.0105)	0.2114 (0.0068)	−0.0133 (0.0091)	1/6	4.46 / 4.83
HiGHS store with GO and the polytope mean against the polytope mean alone	6	0.1981 (0.0105)	0.1252 (0.0414)	+0.0729 (0.0454)	6/6	4.46 / 3.69
HiGHS store with GO and the polytope mean and spread against HiGHS store with GO	6	0.1931 (0.0154)	0.2114 (0.0068)	−0.0184 (0.0159)	2/6	4.23 / 4.83
HiGHS store with GO and the polytope mean and spread against the same without the spread	6	0.1931 (0.0154)	0.1981 (0.0105)	−0.0050 (0.0177)	2/6	4.23 / 4.46
MOMA store with GO appended against GO alone	6	0.2079 (0.0027)	0.2023 (0.0014)	+0.0057 (0.0031)	6/6	4.46 / 5.03
MOMA store with GO appended against HiGHS store with GO	6	0.2079 (0.0027)	0.2114 (0.0068)	−0.0035 (0.0060)	2/6	4.46 / 4.83
HiGHS store with GO at affine decay 0.1 against no decay	6	0.2095 (0.0074)	0.2114 (0.0068)	−0.0019 (0.0080)	4/6	4.66 / 4.83
HiGHS store with GO at affine decay 1.0 against no decay	6	0.2044 (0.0115)	0.2114 (0.0068)	−0.0070 (0.0133)	1/6	4.76 / 4.83
HiGHS store with GO at affine decay 1.0 against decay 0.1	6	0.2044 (0.0115)	0.2095 (0.0074)	−0.0051 (0.0133)	2/6	4.76 / 4.66
HiGHS and MOMA stores with GO against HiGHS store with GO	6	0.2159 (0.0067)	0.2114 (0.0068)	+0.0044 (0.0028)	5/6	4.73 / 4.83
HiGHS and MOMA stores with GO against MOMA store with GO	6	0.2159 (0.0067)	0.2079 (0.0027)	+0.0079 (0.0061)	5/6	4.73 / 4.46
GO multi-hot in the affine slot against the control, twelve seeds	12	0.2029 (0.0019)	0.0759 (0.0547)	+0.1270 (0.0544)	12/12	4.99 / 4.06
HiGHS store with GO appended against GO alone, twelve seeds	12	0.2144 (0.0080)	0.2029 (0.0019)	+0.0115 (0.0085)	11/12	4.98 / 4.99
HiGHS and MOMA stores with GO against HiGHS store with GO, twelve seeds	12	0.2173 (0.0051)	0.2144 (0.0080)	+0.0029 (0.0067)	8/12	4.94 / 4.98

Table 5: Table 1, continued (part 5 of 7).

arm against control	seeds	arm (sd)	control (sd)	difference (sd)	higher	top-50
three families, GO with both flux programs and the measured pool, against GO with both flux programs	6	0.2183 (0.0022)	0.2159 (0.0067)	+0.0024 (0.0076)	3/6	4.63 / 4.73
three families against the pool with GO and the HiGHS vector	6	0.2183 (0.0022)	0.2082 (0.0089)	+0.0101 (0.0089)	5/6	4.63 / 4.66
minimum-norm flux store against the HiGHS parsimonious store	6	0.0931 (0.0131)	0.1302 (0.0335)	−0.0371 (0.0425)	1/6	4.59 / 4.46
minimum-norm flux store against the control	6	0.0931 (0.0131)	0.0582 (0.0746)	+0.0350 (0.0757)	4/6	4.59 / 3.39
minimum-norm flux store with GO against HiGHS store with GO	6	0.2044 (0.0048)	0.2114 (0.0068)	−0.0070 (0.0065)	0/6	4.36 / 4.83
minimum-norm flux store with GO against GO alone	6	0.2044 (0.0048)	0.2023 (0.0014)	+0.0022 (0.0045)	4/6	4.36 / 5.03
measured amino-acid pool alone in the affine against the control	6	0.1947 (0.0288)	0.0582 (0.0746)	+0.1366 (0.0884)	6/6	2.53 / 3.39
measured amino-acid pool alone in the affine against GO alone	6	0.1947 (0.0288)	0.2023 (0.0014)	−0.0075 (0.0298)	4/6	2.53 / 5.03
measured amino-acid pool through a hidden layer of 64 against the pool in the linear affine	6	0.2399 (0.0479)	0.1947 (0.0288)	+0.0451 (0.0468)	5/6	4.03 / 2.53
measured amino-acid pool through a hidden layer of 64 against the control	6	0.2399 (0.0479)	0.0582 (0.0746)	+0.1817 (0.0991)	6/6	4.03 / 3.39
measured amino-acid pool with GO against GO alone	6	0.2082 (0.0024)	0.2023 (0.0014)	+0.0059 (0.0023)	6/6	4.89 / 5.03
measured amino-acid pool with GO and the HiGHS vector against GO with the HiGHS vector	6	0.2082 (0.0089)	0.2114 (0.0068)	−0.0032 (0.0136)	2/6	4.66 / 4.83
measured amino-acid pool with GO and the HiGHS vector against the pool with GO	6	0.2082 (0.0089)	0.2082 (0.0024)	+0.0000 (0.0071)	3/6	4.66 / 4.89
GO with the measured pool through a hidden layer of 64 against the pool alone through the same layer	6	0.2059 (0.0063)	0.2399 (0.0479)	−0.0340 (0.0523)	1/6	4.59 / 4.03
GO with the measured pool through a hidden layer of 64 against the same columns in the linear affine	6	0.2059 (0.0063)	0.2082 (0.0024)	−0.0023 (0.0083)	3/6	4.59 / 4.89
GO with the measured pool through a hidden layer of 64 against GO alone	6	0.2059 (0.0063)	0.2023 (0.0014)	+0.0036 (0.0066)	3/6	4.59 / 5.03
measured amino-acid pool through a hidden layer of 64 against the control, twelve seeds	12	0.2537 (0.0414)	0.0759 (0.0547)	+0.1777 (0.0688)	12/12	4.24 / 4.06
measured amino-acid pool through a hidden layer of 64 against GO alone, twelve seeds	12	0.2537 (0.0414)	0.2029 (0.0019)	+0.0507 (0.0416)	11/12	4.24 / 4.99

Table 6: Table 1, continued (part 6 of 7).

arm against control	seeds	arm (sd)	control (sd)	difference (sd)	higher	top-50
measured amino-acid pool through a hidden layer of 256 against the layer of 64	6	0.2788 (0.0234)	0.2399 (0.0479)	+0.0389 (0.0510)	5/6	4.69 / 4.03
measured amino-acid pool through a hidden layer of 256 against the pool in the linear affine	6	0.2788 (0.0234)	0.1947 (0.0288)	+0.0840 (0.0350)	6/6	4.69 / 2.53
three families through a hidden layer of 64 against the same columns in the linear affine	6	0.2165 (0.0089)	0.2183 (0.0022)	−0.0018 (0.0085)	2/6	4.53 / 4.63
three families through a hidden layer of 64 against GO with the pool through the same layer	6	0.2165 (0.0089)	0.2059 (0.0063)	+0.0106 (0.0142)	5/6	4.53 / 4.59
three families through a hidden layer of 64 against the pool alone through the same layer	6	0.2165 (0.0089)	0.2399 (0.0479)	−0.0234 (0.0462)	1/6	4.53 / 4.03
measured amino-acid pool through a hidden layer of 64 with a budget of 400 epochs against the budget of 200	6	0.2399 (0.0479)	0.2399 (0.0479)	+0.0000 (0.0000)	0/6	4.03 / 4.03
pool with the HiGHS vector through a hidden layer of 64 against the pool alone through the same layer	6	0.1862 (0.0350)	0.2399 (0.0479)	−0.0536 (0.0583)	1/6	2.90 / 4.03
pool with the HiGHS vector through a hidden layer of 64 against the HiGHS store alone	6	0.1862 (0.0350)	0.1302 (0.0335)	+0.0560 (0.0244)	6/6	2.90 / 4.46
measured amino-acid pool through a hidden layer of 512 against the layer of 256	6	0.2425 (0.0476)	0.2788 (0.0234)	−0.0363 (0.0588)	1/6	4.36 / 4.69
measured amino-acid pool through a hidden layer of 512 against the layer of 64	6	0.2425 (0.0476)	0.2399 (0.0479)	+0.0027 (0.0724)	3/6	4.36 / 4.03
measured amino-acid pool through a hidden layer of 128 against the layer of 64	6	0.2622 (0.0283)	0.2399 (0.0479)	+0.0224 (0.0688)	4/6	4.53 / 4.03
measured amino-acid pool through a hidden layer of 128 against the layer of 256	6	0.2622 (0.0283)	0.2788 (0.0234)	−0.0166 (0.0419)	1/6	4.53 / 4.69
measured amino-acid pool through a hidden layer of 256 against the layer of 64, twelve seeds	12	0.2781 (0.0250)	0.2537 (0.0414)	+0.0244 (0.0409)	9/12	4.61 / 4.24
measured amino-acid pool through a hidden layer of 256 against GO alone, twelve seeds	12	0.2781 (0.0250)	0.2029 (0.0019)	+0.0751 (0.0255)	12/12	4.61 / 4.99
measured amino-acid pool through a hidden layer of 256 against the control, twelve seeds	12	0.2781 (0.0250)	0.0759 (0.0547)	+0.2021 (0.0470)	12/12	4.61 / 4.06
measured amino-acid pool through a hidden layer of 256 with a budget of 400 epochs against the budget of 200	6	0.2788 (0.0234)	0.2788 (0.0234)	+0.0000 (0.0000)	0/6	4.69 / 4.69
two-slot head, GO linear beside the pool through 64, against the same columns through one shared layer of 64	6	0.2241 (0.0028)	0.2059 (0.0063)	+0.0182 (0.0069)	6/6	5.66 / 4.59
two-slot head, GO linear beside the pool through 64, against the pool alone through 64	6	0.2241 (0.0028)	0.2399 (0.0479)	−0.0157 (0.0488)	2/6	5.66 / 4.03

Table 7: Table 1, continued (part 7 of 7).

arm against control	seeds	arm (sd)	control (sd)	difference (sd)	higher	top-50
two-slot head, GO linear beside the pool through 64, against the same columns in the linear affine	6	0.2241 (0.0028)	0.2082 (0.0024)	+0.0159 (0.0029)	6/6	5.66 / 4.89
two-slot head, GO linear beside the pool through 64, against GO alone	6	0.2241 (0.0028)	0.2023 (0.0014)	+0.0219 (0.0027)	6/6	5.66 / 5.03
two-slot head, GO linear beside the pool through 256, against the pool alone through 256	6	0.2248 (0.0046)	0.2788 (0.0234)	−0.0540 (0.0204)	0/6	4.99 / 4.69
two-slot head, GO linear beside the pool through 256, against the two slots at 64	6	0.2248 (0.0046)	0.2241 (0.0028)	+0.0006 (0.0039)	4/6	4.99 / 5.66
two-slot head, GO linear beside the pool through 256, against GO alone	6	0.2248 (0.0046)	0.2023 (0.0014)	+0.0225 (0.0048)	6/6	4.99 / 5.03
three families with two slots, GO and both flux programs linear beside the pool through 256, against the same columns in the linear affine	6	0.2392 (0.0033)	0.2183 (0.0022)	+0.0209 (0.0042)	6/6	5.32 / 4.63
three families with two slots against the same columns through one shared layer of 64	6	0.2392 (0.0033)	0.2165 (0.0089)	+0.0227 (0.0104)	6/6	5.32 / 4.53
three families with two slots against GO linear beside the pool through 256	6	0.2392 (0.0033)	0.2248 (0.0046)	+0.0144 (0.0043)	6/6	5.32 / 4.99
three families with two slots against the pool alone through 256	6	0.2392 (0.0033)	0.2788 (0.0234)	−0.0396 (0.0217)	0/6	5.32 / 4.69
two-slot head, GO linear beside the pool through 128, against the pool alone through 128	6	0.2252 (0.0052)	0.2622 (0.0283)	−0.0370 (0.0263)	1/6	5.49 / 4.53
two-slot head, GO linear beside the pool through 128, against the two slots at 64	6	0.2252 (0.0052)	0.2241 (0.0028)	+0.0010 (0.0049)	3/6	5.49 / 5.66
two-slot head, GO linear beside the pool through 128, against the two slots at 256	6	0.2252 (0.0052)	0.2248 (0.0046)	+0.0004 (0.0038)	4/6	5.49 / 4.99
measured amino-acid pool through a hidden layer of 256 with transformer dropout 0.3 against without	6	0.2787 (0.0183)	0.2788 (0.0234)	−0.0001 (0.0279)	3/6	4.56 / 4.69
measured amino-acid pool through a hidden layer of 256 read standardized against log1p	6	0.1820 (0.0170)	0.2788 (0.0234)	−0.0968 (0.0226)	0/6	3.99 / 4.69
three families with two slots against the pool alone through 256, twelve seeds	6	0.2392 (0.0033)	0.2781 (0.0250)	−0.0396 (0.0217)	0/6	5.32 / 4.61
three families with two slots against GO alone, twelve seeds	6	0.2392 (0.0033)	0.2029 (0.0019)	+0.0369 (0.0043)	6/6	5.32 / 4.99
three families with two slots against the HiGHS store with GO, twelve seeds	6	0.2392 (0.0033)	0.2144 (0.0080)	+0.0277 (0.0082)	6/6	5.32 / 4.98

Table 8: The pinned-test Spearman of the 027 rounds on the same split and statistic, read by `target_transfer_readout.py` from 027's result files: the mean over three seeds for a trained arm, the validation-selected cell for the linear ladder, and the Merzbacher comparison's RandomForest.

comparator	Spearman
betaxanthin	
027 flux precursor readout (wave 8), three seeds	0.1313
027 context-only control of the precursor readout (wave 9), three seeds	0.1152
cosine kNN over ProtT5 of the deleted gene (k 25, job 1909)	0.1446
FCL RandomForest (Merzbacher 2025)	0.0391

3 Physically constrained stores on the strain, and the store read without the backbone ✗

The module so far carries mass balance and nothing else: one parsimonious vector per deletion. On the essentiality endpoint the store ladder (the essentiality document's round 113) built the same vector under enzyme ceilings, sampled turnover, GECKO pools and a minimization of metabolic adjustment, and none moved the score beyond the seed spread. None of those programs had been run on the target model, the cassette strain with its markers deleted on YPD, which covers every GEM gene rather than the labeled ones. On 2026-09-23 they were, on CPU beside the packs, with reads that need no card.

The second law as fixed directions admits no growth. The direction rule of the essentiality study's round 114 (a reaction whose standard energy on the union-external route is at least T below zero loses its reverse part, at least T above it its forward part; unpriced reactions, the cassette's included, keep the GEM box) leaves the parent strain infeasible at both thresholds, as it had on the essentiality model. On the cassette strain the 10 kJ/mol rule fixes 1,754 directions forward and 1,068 reverse, and 97 of them run against the parsimonious wild-type flux; at 30 kJ/mol it fixes 1,071 and 905 with 69 conflicts. No store is written for an infeasible parent (a store of zero rows would be read as every deletion lethal), and the outcome is recorded (`results/target_physics_store_betaxanthin_sign10.json`, `_sign30`). Standard energies at 1 M cannot be imposed as directions on this model; a thermodynamically consistent program needs the concentrations as variables, which is what the driving-force store below does with a linear program.

The stores that do exist. The linear MOMA store on the cassette strain (1,155 genes, 27 s on eight processes) has 107 lethal deletions against the parsimonious store's 93, moves the growth ratio by more than 0.01 on 59 genes, and moves the $\log(1+|v|)$ vector on 1,056 genes with a median relative distance of 0.39 to the parsimonious deletion vector. The parsimonious store rebuilt on the HiGHS path keeps the 93 lethal deletions, moves 17 ratios, and sits at a median distance of 0.015 from the GLPK build. The max-min driving-force store (`target_driving_force_store.py`) solves, per deletion, for the metabolite concentrations between 1 μM and 10 mM that maximize the smallest driving force over the reactions the parsimonious flux carries, with transport and proton-translocating reactions unconstrained and, in the capped variant, reactions priced above 100 kJ/mol set aside; the wild type's max-min driving force is -1.16 kJ/mol over 172 constrained reactions with 32 binding, 1,062 of 1,062 viable deletions have a negative optimum, and 451 change the concentration vector. The sampled store (`target_sampled_flux_store.py`) draws six OptGP samples of each deletion's cone at 99 percent of its optimum, 1,155 deletions in 6 h on twelve processes, and keeps the mean over draws and, in a second store, the per-reaction spread.

Why the MOMA store differs so much from the parsimonious one: the GLPK build is degenerate (Table 13, from `store_degeneracy.py`). Among the 1,031 deletions that leave growth unchanged, the GLPK store's vector differs from its own wild type's for 1,025, at a median relative distance of 0.38 in the encoding the affine reads; the HiGHS rebuild differs for 428 (median 0.0, ninetieth percentile 0.39) and the MOMA store, anchored to one wild-type vector, for 18 (the prototroph's stores are in the sibling document). So the parsimonious optimum has alternative vertices for a large share of no-effect deletions under either solver, and the GLPK path lands on a different one almost every time. Hypothesis (untested): the vertex noise is what erased the growth block's signal in round 125, and round 139 is the transformer read of the same store without it.

A store without vertices. A faster or better linear solver would pick a third set of vertices of the same degenerate optimum, so the vertex noise is removed by the program, not the solver: the `l2` variant of `target_physics_store.py` solves the growth LP, then minimizes $\|v\|_2^2$ subject to $Sv = 0$, the knocked-out box and $v_{biomass}$ held at the LP optimum (less 10^{-6} relative), one interior-point QP per gene (Clarabel; HiGHS's active-set QP returns a solve error on this 2,831 by 4,177 model). The squared norm is strictly convex on the optimal face, so each deletion has one solution whatever the path to it. The 1,155 genes solve in 49 s with every solve converged, the source's 93 lethals and the HiGHS rebuild's 17 moved growth ratios reproduced. The minimum-norm point spreads flux where the ℓ_1 point concentrates it: the wild type carries 996 reactions above 10^{-6} against the parsimonious solution's 426, at an ℓ_1 norm of 452 against 426. Its degeneracy row (Table 13) reads 427 no-effect deletions differing from the wild type by more than the script's 10^{-6} threshold, but at a median relative distance of 10^{-8} and a ninetieth percentile of 3×10^{-5}, against 0.39 for the HiGHS store and 0.38 for the GLPK one: interior-point tolerance, not vertex choice. On the ladder (Table 14) the minimum-norm store reads 0.068 by ridge, 0.126 by nearest neighbors and 0.133 by forest, against the HiGHS ℓ_1 store's 0.098, 0.195 and 0.117: less rank under the two distance models, a little more under the forest, and the weakest calls of any flux store (0.40 to 2.00). Appended to GO it reads 0.068, 0.208 and 0.173 against the ℓ_1 vector's 0.123, 0.198 and 0.204: the highest nearest-neighbor read of any genotype-side representation, and less by ridge and forest. So removing the vertex noise does not add rank on its own; a plausible reason, unverified, is that the ℓ_1 point's sparsity pattern, which reactions carry flux at all, is what the cosine neighborhoods read, and the minimum-norm point spreads that pattern over twice the reactions. The transformer's affine answers the same way (rounds 181, 182): the store alone reads 0.093 against the HiGHS store's 0.130 on the same seeds, under it on five of six, and appended to GO it reads 0.204 against the HiGHS vector's 0.211, under it on six of six and within 0.002 of GO alone. The affine prefers the sparse point to the unique one, and the flux side's gain beside the annotation, +0.012 on twelve seeds for the ℓ_1 vector, is not carried by the ℓ_2 point at all.

The store read without the backbone (Table 14; `target_store_baselines.py`, the 027 ladder's split, standardization and selection, with a random forest added, the Flux Cone Learning form on one point per deletion). Two things frame the numbers. The screen is genome-wide and the stores cover the 1,155 GEM genes, so 3,255 of the 3,494 betaxanthin training genotypes read the wild-type row and every store baseline learns from about 240 in-model deletions, while the pinned test set is inside the model (628 of 629). And a gene's row is one vector with no graph, no sequence and no pathway. Two of the ladder's reads were fragile to this degeneracy and are made exact: a nearest-neighbor query on the wild-type row has about two thousand identical training rows, so the k neighbors taken by sort order were an arbitrary subset that changed between runs, and every training row tied with the k-th similarity is now averaged in; predictions within 10^{-9} count as one rank; and the validation criterion is rounded to 10^{-6} before the grid is chosen. Two consecutive runs then agree to every printed digit. On betaxanthin the GLPK growth block carries nothing (ridge -0.006, kNN -0.004, forest $+0.009$); the whole GLPK vector, the one round 117 read, carries 0.097 by kNN; the HiGHS rebuild of the growth program carries 0.195 by kNN with a top-50 enrichment of 3.39, the highest rank of any flux-side read on this split (the best single transformer arm is 0.121, the best six-seed ensemble 0.153, the cosine kNN over ProtT5 0.145, reproduced here at 0.144); the MOMA store 0.135 by ridge; the six flux statistics of the parsimonious build 0.145 by kNN; the capped driving-force concentrations 0.152 by ridge with a top-50 of 3.39, the metabolite side carrying rank linearly. The mean of six samples of each deletion's flux cone at 99 percent of optimum carries 0.007 by ridge, 0.064 by kNN and 0.102 by forest, below the parsimonious point it is sampled around, as the sampled stores sat on essentiality (the essentiality document's round 87). The forest is not the best model on any flux representation, so the Flux Cone

Learning form does not rescue a point representation. Rank and call separate as they did in every transformer round: ProtT5 gives the best top-50 (4.59 alone, 4.99 with the GLPK store under a forest), the clean flux store the best rank.

The region, not its point. The max-min program returns one concentration vector, but the second law does not single it out: every log-concentration vector within the bounds whose carried priced reactions all run downhill by at least the optimum is as consistent, and that set is a polytope in $\ln c$. `target_concentration_polytope_store.py` samples it per deletion, the region slackened by 1 kJ/mol below B^* so it has volume where several reactions bind, by coordinate hit-and-run from the point program's solution (a random direction stalls when the start sits on the box, as the wild type's does; one metabolite per step does not), 20,000 burn-in steps, six draws 2,000 steps apart, the metabolites in no constrained reaction uniform in the bounds. The chain is seeded from a hash of the polytope rather than the row, so two deletions with the same region read the same row; seeded by row, every one of the 1,062 viable rows differed from the wild type's, a sampling-noise degeneracy of the GLPK kind, and under the polytope hash 455 do, beside the point store's 451. The wild type's region constrains 234 of the 2,803 variable metabolites, and over those the median spread of $\ln c$ across draws is 2.18 against 2.63 for a metabolite the constraints never touch (uniform over four decades), so at this cap the second law narrows a typical constrained metabolite by little and the information sits in the few it pins; the mean over draws sits a median 0.12 from the point in relative distance. Two stores, the mean concentration over draws with the point store's statistics and the count of constrained metabolites, and that mean with the per-metabolite spread of $\ln c$ appended. On the ladder the polytope mean reads 0.136 by ridge, 0.127 by kNN and 0.126 by forest, the one flux-side representation that every model reads above 0.12 (the point gives ridge 0.152 but kNN 0.072 and forest 0.098; the HiGHS flux vector gives kNN 0.195 but forest 0.117), and the spread store 0.096, 0.138 and 0.128, so the width of the region carries rank the mean does not under nearest neighbors. Round 147 (w164) is the transformer read of the polytope mean, paired against the point.

The annotation-only family, on the same split (Table 12; `target_go_models.py`). The essentiality study read DCell's ontology hierarchy and a graph convolution over the GO multi-hot beside the transformer; neither had a target head, so both are given one here on the ladder's split: DCell's hierarchy in the essentiality script's shared-input form with a linear head on the root's state and auxiliary regressions, and a perceptron with one hidden layer of 256 over the gene's propagated GO multi-hot (2,341 of 2,366 columns with training support), which is what the GO graph convolution reduces to on one deletion per instance; the same perceptron over the HiGHS parsimonious store row, over ProtT5 and over their concatenations separates the annotation, the sequence and the flux families on one model. Full batch, Adam at 10^{-3}, a masked squared error on standardized targets, 200 epochs, the epoch selected on validation per-feature Pearson, six seeds. The GO perceptron reads 0.200 (sd 0.002 over seeds) with a top-50 enrichment of 4.79, selected at epoch 21 or 22 on every seed and trained in three seconds a seed: above every transformer arm (the best single 0.121, the best six-seed ensemble 0.153), above the HiGHS store's nearest neighbors (0.195) and with a call at the best transformer arms' level. The store row through the same perceptron reads 0.119 with a top-50 of 2.00, ProtT5 0.103 and 4.13, so the two inputs the transformer arms read, ProtT5 with the pathway multi-hot, are the weaker two of the three families on this target and the pathway multi-hot is a weaker annotation than propagated GO. Concatenation adds no rank: GO with the store 0.198 (sd 0.041), GO with ProtT5 0.173 with the best calls (4.99 single, 5.39 ensembled), all three 0.175 and 5.03. DCell's hierarchy over the same columns reads 0.049 (sd 0.043), selected at epochs 8 to 13, under the perceptron on every seed: the one-subsystem-per-term wiring costs three quarters of what the flat annotation carries. DCell's own input, a weight vector per (gene, term) annotation pair in place of the shared one, reads 0.028 (sd 0.058) with a top-50 of 1.26, and 0.073 with 1.40 when the six seeds are averaged; the seeds select anywhere from epoch 0 to 93 and read -0.057 to 0.115. Under a split by gene a held-out gene's columns receive no gradient, so the published network places an unseen gene through its random initial columns; that the read is not exactly zero is consistent with the trained subsystems responding to how many terms the gene lights rather than to which (a hypothesis, untested). The shared input is what lets the hierarchy read a gene it has not seen, and with it the hierarchy still reads a quarter of the flat perceptron. Width 20, the other point of the essentiality study's tune grid, lifts the shared hierarchy to 0.069 (sd 0.036) with a top-50 of 3.33 (0.066 and 3.79 ensembled), at 13 to 24 minutes a seed against 7 to 8 at width 8, and leaves the published input at 0.041 (sd 0.046), three of its six seeds never reaching a validation peak in 200 epochs; both stay at a third of the perceptron or under. DCell's published sizing, max(20, 0.3 times the genes under the term), is out of reach for this implementation, which pads every term to the widest: the root's 1,650 units would put the edge tensor past 30 GB. Two things bound the reading. GO annotation is curated in part from deletion phenotypes, respiratory function among them, and the screen's top hits are respiratory (the 027 read), so an annotation input can carry the target through the literature the target itself produced; the split holds out genes, not annotations. And the transformer arms were never given this input: the annotation the backbone reads is the pathway multi-hot, and rounds 150 and 151 (w167, w168) put the GO multi-hot in the affine slot, alone and appended to the HiGHS store, over every annotated gene, the interface every other input has entered through. The same two inputs read by the ladder's three models (Table 14) fix what the transformer adds to them: the GO multi-hot reads 0.076 by ridge, 0.179 by nearest neighbors and 0.187 by forest with a top-50 enrichment of 5.39, the highest call of any genotype-side read on this split; GO with the HiGHS vector appended reads 0.123, 0.198 and 0.204 with a top-50 of 4.79. The forest over the 6,539 columns round 151's affine reads sits 0.008 under that round's 0.211, the size of the round's own seed spread, with the higher call, so on this input the transformer's rank is what a backbone-free model of the same columns reaches, and averaging its six seeds adds 0.001.

The measured side enters the same ladder. The strain's own Mülleder amino-acid pool, nineteen concentrations in mM as an affine store over the 4,678 strains the screen measured (a strain it did not measure reads the loader's reference, the population robust mean per amino acid), reads 0.149 by ridge, 0.285 by nearest neighbors and 0.280 by forest, with top-50 enrichments of 3.99, 4.19 and 4.79: nineteen columns carry more rank than any genotype-side representation and more than any transformer arm (0.216, round 171). Appended to GO the pool reads 0.102, 0.174 and 0.264 with a forest call of 5.99; appended to GO with the HiGHS vector, 0.156, 0.195 and 0.247 with a forest call of 6.19, the highest call of any read on this split. The ridge and the nearest-neighbor reads fall when the annotation's 2,362 columns join the nineteen while the forest, which selects columns, keeps 0.26 and gains the call; a plausible reason, unverified, is that the two distance-based models weigh every standardized column alike, so the pool is then a hundredth of the input. This read is phenotype to phenotype: the pool is a measurement of the same strain in a different screen, on minimal medium, not the strain's sequence, and the split by gene holds because the pool row is never the betaxanthin target. What it fixes is that the amino-acid data carries betaxanthin signal beyond the gene's identity, at a rank above the sequence, the annotation and every flux store. The transformer's affine reads the same columns linearly at 0.195 alone and 0.208 beside GO (rounds 179, 175), above the ridge and under the distance models; through a hidden layer of 64 the pool alone reads 0.240 with a six-seed ensemble of 0.292, above the nearest neighbors (round 180), and through a layer of 256 it reads 0.279 (round 187).

The same read without the transformer, on CPU, sets what the columns give a perceptron alone (Table 9; the GO-models script with the pool as its store block, one hidden width per pass, six seeds, at most 14 s a seed). The pool alone climbs with width, 0.178, 0.217, 0.219 and 0.236 at 16, 64, 256 and 1024, the last with the tightest band (sd 0.007) and a six-seed ensemble of 0.240; GO appended reads 0.237, 0.232, 0.238 and 0.240 at the same widths, flat in width, sd 0.003 at 1024 with an ensemble of 0.241 and the best call of the sweep (top-50 5.16), selecting at epochs 15 to 16. Two things differ from the transformer's slot. The annotation adds to the pool here at every width, 0.02 over the pool alone at 1024 and 0.04 over the GO perceptron alone (0.200), where inside the transformer's hidden layer it took the read back to the annotation's level (round 185); and the transformer's read of the pool alone through 256, 0.279, sits 0.06 above the standalone perceptron at the same width and 0.04 above its best width. The two trainings are not one setting under two heads, since the perceptron standardizes the columns, trains full batch and selects on a smoothed validation curve, so which of those carries the gap is not measured; the plain reading is that the columns give a perceptron 0.24 and the transformer's hidden slot 0.28, and that the annotation's effect on the pool's read depends on the head it enters.

Two of those differences are now measured and neither accounts for the gap. Read under the transformer's own encoding, the columns as $\log(1+v)$ rather than standardized, the perceptron falls to 0.158 at width 256 and 0.173 at 1,024, and a 1,000-epoch budget moves the wider setting only to 0.177, so the raw read is converged and the encoding costs the perceptron 0.06. Its best over both encodings is therefore 0.236, and the transformer reads the same nineteen columns at 0.279. What remains between them is the backbone, the batch of 32 against full batch, the continuous ranked probability score against a masked squared error, and the smoothed selection; which of those carries the 0.043 is not measured, but the budget and the encoding do not.

The two-slot form goes the other way, and this is what the transformer's rounds 197 to 199 reproduce. When the annotation's columns enter through the store, as they do in the transformer's two-slot head, the encoding decides what the linear slot gives: under the raw encoding that arm reads 0.243 for GO with both flux programs and the pool and 0.237 for GO with the pool, and at 1,000 epochs 0.242 and 0.237, where standardizing the same columns reads 0.106 and 0.179. Standardizing a binary multi-hot is what costs it, and the transformer, which reads its store as $\log(1+v)$, sits with the raw column: its three-families two-slot head reads 0.239 against the perceptron's 0.242, within 0.003. So on the pool alone the two heads differ by 0.043 with nothing left to explain it but the backbone and the fit; on the two-slot form over the same concatenated store they agree to a thousandth. A head with two slots, the GO multi-hot through a linear map and the pool through the hidden layer, summed before the output so the two never share a hidden unit, reads 0.260 at width 256 and 0.265 at 1,024 (sd 0.006, ensemble 0.266, top-50 5.36, 23,872 parameters): above the shared layer over the same columns (0.240) and the pool alone (0.236) at every width, the highest single-seed read of any perceptron in the study and the best call of the sweep. Depth costs the pool alone and not the two-slot head: at width 1,024 a second and a third hidden layer take the pool alone to 0.185 and 0.195 and the shared layer to 0.247 and 0.243, while the two slots hold at 0.262 and 0.258. So what the annotation does to the pool's read is set by whether the two share a hidden layer: sharing costs the pool 0.02 to 0.03 in the perceptron and 0.03 in the transformer, and a linear slot beside the pool's hidden layer adds 0.03 to the pool alone in the perceptron. That head is the transformer arm the study has not built. The ladder agrees with the transformer on the flux side: the pool with the HiGHS vector appended on the pool's own genes (round 191's store) reads 0.095, 0.153 and 0.220 by ridge, nearest neighbors and forest against the pool alone's 0.149, 0.285 and 0.280, under it on every model and by 0.13 for the neighbors (Table 14). The second amino-acid screen read the same way, Cooper's seventeen relative peaks of the strain as a store on the pool's construction (4,313 strains, 8.5 percent of cells missing and carrying the reference 1.0; 466 of the ladder's rows are strains Cooper did not measure and carry the reference row), reads -0.001, 0.094 and 0.113 by ridge, nearest neighbors and forest with calls of 0.6 to 1.8: a third of Mülleder's pool on every model. Cooper's own duplicate identifiers agree at a Spearman of 0.06 (the sibling document's Section 4), so the columns carry little of the strain to begin with. Through

the perceptron (Table 10; the pool sweep's arms with Cooper's store) the peaks alone read 0.131 at width 256 with calls at chance (top-50 1.03), the forest's level, and GO linear beside them 0.233, under GO beside the pool (0.260) by 0.027. Appended to the pool (36 measured columns, store `YPD_bx_aapoolcooper`) the peaks read 0.245 alone against the pool's 0.219 at the same width, +0.026 on three seeds of six, with the pool's spread of 0.042 brought to 0.010: two seeds of the pool alone select at epoch 18 to 21 under 0.17, and with Cooper's columns beside them neither does, so at this width the second screen steadies the first more than it adds to it, and against the pool alone at its best width (0.236 at 1,024) the gain is 0.009. In the two-slot head the peaks add 0.004 to the pool on five seeds of six (0.264 against 0.260, ensembles 0.265 and 0.266): within the band, and not a cost. Cooper's peaks are read as a substitute for the pool rather than an addition to it: their transformer read through 256, alone, is round 211 (w230), and the appended form (w228) was withdrawn unstarted.

Table 9: The strain's measured amino-acid pool (nineteen columns, log1p then standardized on the training rows) alone and with the propagated GO multi-hot, through one ReLU hidden layer of the stated width and a linear head, on the ladder's split, full batch, Adam at 10^{-3}, 200 epochs, the epoch selected by a five-epoch moving average of the validation per-feature Pearson, on CPU. The two-slot rows read the GO multi-hot through a linear map and the pool through the hidden layers, summed before the output, so the two never share a hidden unit. Columns: hidden width, the number of hidden layers, parameters, the single-seed pinned-test Spearman with the standard deviation over six seeds and, after the slash, the top-50 enrichment; the six-seed ensemble scored the same way; the range of selected epochs. Written by `target_notes_tex_tables.py` from `results/go_models_betaxanthin_pool_*.json` (`target_go_models.py -store YPD_bx_aapool -mlp-hidden H -mlp-depth D`).

model	width	depth	parameters	single seed (sd) / top-50	ensemble	epochs
perceptron over the measured pool	16	1	337	+0.178 (0.012) / 4.49	+0.184 / 4.59	110 to 198
perceptron over GO with the measured pool	16	1	38,193	+0.237 (0.004) / 5.23	+0.239 / 5.39	45 to 59
perceptron over the measured pool	64	1	1,345	+0.217 (0.012) / 4.33	+0.227 / 4.19	62 to 198
perceptron over GO with the measured pool	64	1	152,769	+0.232 (0.018) / 4.99	+0.236 / 4.79	30 to 107
perceptron over the measured pool	256	1	5,377	+0.219 (0.042) / 4.36	+0.245 / 4.39	18 to 184
perceptron over GO with the measured pool	256	1	611,073	+0.238 (0.003) / 4.96	+0.239 / 4.99	20 to 21
perceptron over the measured pool	1024	1	21,505	+0.236 (0.007) / 4.36	+0.240 / 4.39	52 to 87
perceptron over GO with the measured pool	1024	1	2,444,289	+0.240 (0.003) / 5.16	+0.241 / 4.99	15 to 16
GO linear beside the pool's perceptron (two slots)	256	1	7,744	+0.260 (0.008) / 5.09	+0.266 / 5.39	64 to 165
GO linear beside the pool's perceptron (two slots)	1024	1	23,872	+0.265 (0.006) / 5.36	+0.266 / 5.39	72 to 105
perceptron over the measured pool	1024	2	1,071,105	+0.185 (0.021) / 4.49	+0.195 / 4.59	16 to 53
perceptron over GO with the measured pool	1024	2	3,493,889	+0.247 (0.003) / 5.39	+0.255 / 5.99	8 to 13
GO linear beside the pool's perceptron (two slots)	1024	2	1,073,472	+0.262 (0.006) / 5.46	+0.265 / 5.59	30 to 73
perceptron over the measured pool	1024	3	2,120,705	+0.195 (0.033) / 4.73	+0.196 / 4.79	14 to 24
perceptron over GO with the measured pool	1024	3	4,543,489	+0.243 (0.005) / 5.46	+0.244 / 5.59	11 to 13
GO linear beside the pool's perceptron (two slots)	1024	3	2,123,072	+0.257 (0.008) / 5.29	+0.261 / 5.59	26 to 33

Table 10: Measured stores through a perceptron predicting the betaxanthin score, on CPU: the measured pool (nineteen columns), Cooper's seventeen relative peaks, and the pool with Cooper's peaks appended (36 columns, store YPD_bx_aapoolcooper from `target_store_concat.py`), each log1p then standardized on the training rows, alone, with the propagated GO multi-hot through the same hidden layer, and with GO through a linear map beside the store's hidden layer (two slots), one ReLU hidden layer of the stated width and a linear head, on the ladder's split, full batch, Adam at 10^{-3}, 200 epochs, the epoch selected by a five-epoch moving average of the validation per-feature Pearson. Columns: the store, the arm, hidden width, parameters, the single-seed pinned-test Spearman with the standard deviation over six seeds and, after the slash, the top-50 enrichment; the six-seed ensemble scored the same way, the range of selected epochs. Written by `target_notes_tex_tables.py` from `results/go_models_betaxanthin_*.json` (`target_go_models.py -store <store> -mlp-hidden H`).

store	model	width	parameters	single seed (sd)	ensemble	epochs
pool	perceptron over the store	256	5,377	+0.219 (0.042) / 4.36	+0.245 / 4.39	18 to 184
pool	perceptron over GO with the store	256	611,073	+0.238 (0.003) / 4.96	+0.239 / 4.99	20 to 21
pool	GO linear beside the store's perceptron (two slots)	256	7,744	+0.260 (0.008) / 5.09	+0.266 / 5.39	64 to 165
Cooper	perceptron over the store	256	4,865	+0.131 (0.007) / 1.03	+0.143 / 1.00	69 to 198
Cooper	perceptron over GO with the store	256	610,561	+0.207 (0.003) / 4.63	+0.207 / 4.59	20 to 22
Cooper	GO linear beside the store's perceptron (two slots)	256	7,232	+0.233 (0.011) / 4.46	+0.234 / 4.39	87 to 188
pool + Cooper	perceptron over the store	256	9,729	+0.245 (0.010) / 3.99	+0.255 / 4.19	77 to 144
pool + Cooper	perceptron over GO with the store	256	615,425	+0.241 (0.004) / 4.59	+0.242 / 4.59	21 to 23
pool + Cooper	GO linear beside the store's perceptron (two slots)	256	12,096	+0.264 (0.007) / 4.56	+0.265 / 4.79	66 to 101

3.1 The backbone's head and the slot, separated ✗

A store arm predicts $h(z) + a(s)$, the backbone's context head over the cell graph plus the slot over the gene's store row, and the two can be read apart: each cell is rebuilt from its checkpoint at the selected epoch, the pinned test set is run once as trained and once with the slot replaced by zero, and the difference is the slot (Table 11;

`target_head_decomposition.py`, the rebuilt full prediction agreeing with the reported Spearman to a thousandth on every cell). On the pool through 256 (round 187) the head is a constant on two seeds of six, with a spread of 10^{-8} of the slot's, the transformer blocks' gradient norm at 0.0000 from epoch 20 and the 75 pinned genes with no pool row all receiving one prediction; on the other four the head is live, reads 0.05 to 0.15 on its own, agrees with the slot at 0.04 to 0.14, and adds +0.055, +0.046, +0.047 and −0.020 to the slot's read. Over the six seeds the slot alone reads 0.257 and the backbone adds 0.021. So the arm's 0.279 is the pool through the hidden layer, fitted by this trainer, plus a backbone that adds 0.05 when it survives and nothing when it does not; the 0.043 between the transformer and the perceptron over the same columns (0.236) is 0.02 of fit and 0.02 of a live backbone, and not the graph. With the annotation in the slot the head is a constant on every seed: the GO affine (round 150), the two-slot head (198) and the three families with two slots (199) select at epochs 7 to 15, before the blocks have moved, and their reads are the slot's within two ten-thousandths. The flux affine (139) is the one arm whose head is live on every seed, with the head's spread at the slot's and the two parts agreeing at 0.13; it adds 0.023 there. The pool linear (181) and the pool through 64 (182) add 0.004 and −0.004.

The graph never reached any of these backbones. The 019 regularizer and the 020 hard mask both index the second transformer block, and every cell of this study ran one block: the regularizer's loss is 0.0 at every epoch of all 384 one-block cells through round 200 and nonzero only on the 24 two-block width-90 cells, and the hard mask was never passed. The backbone's inputs were the pathway multi-hot and ProtT5 through unmasked, unregularized attention over all genes. The layer-pattern ladder on the pool through 256 (rounds 212 to 216: one block of width 36 with nine heads, two unmasked blocks, two blocks with the regularizer on the second, two with the hard mask on the second, and three with the hard mask on the second and third) is the first read of the graphs in this study, and the release rule on the pool slot (round 210) asks whether a backbone that escapes first adds on every seed.

Table 11: Each store arm's pinned-test prediction rebuilt from its checkpoint at the selected epoch and split into the backbone's context head $h(z)$ (the affine replaced by zero) and the slot $a(s)$ (the difference), six seeds an arm. Columns: the seeds on which the head is a constant on the pinned genes (its spread under a hundredth of the slot's); the Spearman of the full prediction and of the slot alone against the target, over all six seeds; over the seeds with a live head, the head's own Spearman, the ratio of the head's spread to the slot's, and the Spearman between the two parts; and what the backbone adds, the full read minus the slot's, over all six seeds. Written by `target_notes_tex_tables.py` from `results/target_head_decomposition_*.json` (`target_head_decomposition.py`).

arm	constant head	full	slot	head	sd head/slot	head vs slot	backbone adds
GO affine (round 150)	6 of 6	+0.202	+0.202	–	–	–	-0.000
HiGHS flux affine (round 139)	1 of 6	+0.130	+0.107	+0.060	1.02	+0.14	+0.023
pool, linear slot (round 181)	1 of 6	+0.195	+0.191	+0.104	0.67	+0.04	+0.004
pool through 64 (round 182)	2 of 6	+0.240	+0.244	+0.059	1.30	+0.14	-0.004
pool through 256 (round 187)	2 of 6	+0.279	+0.257	+0.100	0.49	+0.08	+0.021
GO linear beside the pool through 256 (round 198)	6 of 6	+0.225	+0.225	–	–	–	+0.000
three families linear beside the pool through 256 (round 199)	6 of 6	+0.239	+0.239	–	–	–	-0.000

Table 12: DCell's ontology hierarchy (the essentiality study's shared-input form, a linear head on the root's state, auxiliary regressions at alpha 0.3) and a perceptron over the propagated GO multi-hot, on the ladder's split (2,341 GO columns with training support), full batch, Adam at 10^{-3}, 200 epochs, the epoch selected by a five-epoch moving average of the validation per-feature Pearson. Columns: parameters, the single-seed read (the pinned-test Spearman with the standard deviation over seeds and, after the slash, the top-50 enrichment), the six-seed ensemble scored the same way, and the selected epochs. Hierarchy rows are at subsystem width 8 unless marked width 20. Written by `target_notes_tex_tables.py` from `results/go_models_betaxanthin.json` and its `_w20` companion (`target_go_models.py`).

model	parameters	single seed (sd)	ensemble	selected epochs
DCell-style hierarchy, shared gene input, root head	354,615	+0.049 (0.043) / 3.10	+0.046 / 3.00	13, 8, 11, 9, 9, 8
DCell hierarchy, published input (a weight per gene and term), root head	1,972,151	+0.028 (0.058) / 1.26	+0.073 / 1.40	0, 81, 91, 93, 2, 26
perceptron over the GO multi-hot	606,209	+0.200 (0.002) / 4.79	+0.200 / 4.79	21, 21, 21, 22, 21, 21
perceptron over the HiGHS parsimonious store row	1,069,825	+0.119 (0.013) / 2.00	+0.121 / 2.00	16, 13, 9, 13, 31, 20
perceptron over ProtT5	262,657	+0.103 (0.023) / 4.13	+0.107 / 4.79	7, 6, 10, 9, 7, 8
perceptron over GO with the store row	1,675,521	+0.198 (0.041) / 4.86	+0.189 / 4.99	26, 29, 22, 28, 28, 23
perceptron over GO with ProtT5	868,353	+0.173 (0.011) / 4.99	+0.177 / 5.39	4, 8, 4, 8, 13, 13
perceptron over GO, ProtT5 and the store row	1,937,665	+0.175 (0.015) / 5.03	+0.184 / 5.19	11, 11, 11, 10, 11, 19
DCell-style hierarchy, shared gene input, root head (width 20)	1,848,987	+0.069 (0.036) / 3.33	+0.066 / 3.79	4, 2, 3, 14, 6, 5
DCell hierarchy, published input (a weight per gene and term), root head (width 20)	5,892,827	+0.041 (0.046) / 1.70	+0.099 / 2.40	74, 15, 200, 200, 0, 200

Table 13: Among the deletions whose growth ratio is 1, how many carry a vector that differs from the store's own wild-type row (relative ℓ_2 above 10^{-6} in $\log(1 + |v|)$), with the median and ninetieth percentile of that distance. Written by `target_notes_tex_tables.py` from `031-betaxanthin-module/results/store_degeneracy.json` (`store_degeneracy.py`, which reads every target store).

store	genes	no effect	differ	median	p90
parsimonious, cobra pFBA through GLPK	1155	1031	1025	0.383	0.383
parsimonious, HiGHS two-stage	1155	1031	428	0.000	0.390
minimum-norm flux at maximal growth (Clarabel QP)	1155	1031	427	0.000	0.000
linear MOMA	1155	973	18	0.000	0.000

Table 14: Each deleted gene represented by one row of a store ($\log(1 + |v|)$ of the flux columns, standardized on the training genes; a gene the store does not cover reads the wild-type row), fit by the 027 ladder's ridge (rank-swept bilinear for the amino acids) and cosine nearest neighbors, every training row tied at the k-th similarity averaged in, and by a 300-tree random forest, the Flux Cone Learning form on one point. Selected on validation per-feature Pearson rounded to 10^{-6}; the cell is the pinned-test Spearman and, after the slash, the top-50 enrichment. Written by `target_notes_tex_tables.py` from `results/store_baselines_betaxanthin.json` (`target_store_baselines.py`).

representation	columns	ridge	kNN	forest
GLPK parsimonious store (growth, production, total)	8355	-0.004 / 1.80	+0.097 / 1.60	+0.027 / 3.19
GLPK parsimonious store, growth block	4177	-0.006 / 1.00	-0.004 / 0.80	+0.009 / 2.40
HiGHS parsimonious store	4177	+0.098 / 2.00	+0.195 / 3.39	+0.116 / 1.00
minimum-norm flux at maximal growth (Clarabel QP, one solution per deletion)	4177	+0.068 / 0.40	+0.126 / 1.20	+0.133 / 2.00
linear MOMA store	4177	+0.135 / 2.20	+0.088 / 1.40	+0.072 / 1.00
six gene-rule counts	6	+0.059 / 2.00	+0.032 / 1.40	+0.056 / 1.40
flux statistics of the parsimonious build	6	+0.107 / 1.80	+0.145 / 3.00	+0.102 / 0.80
driving-force concentrations, every priced reaction	2834	+0.081 / 0.40	+0.119 / 2.00	+0.063 / 0.60
driving-force concentrations, cap 100 kJ/mol	2834	+0.152 / 3.39	+0.072 / 1.40	+0.098 / 0.80
sampled cone, mean over draws	4177	+0.007 / 1.40	+0.064 / 1.00	+0.102 / 2.20
driving-force polytope, mean concentration over draws	2835	+0.136 / 2.00	+0.127 / 2.00	+0.126 / 2.00
driving-force polytope, mean with the spread of ln c	5666	+0.096 / 2.00	+0.138 / 1.60	+0.128 / 2.20
GO multi-hot (2,362 propagated terms)	2362	+0.076 / 2.00	+0.179 / 4.39	+0.187 / 5.39
HiGHS parsimonious store with GO appended	6539	+0.122 / 2.00	+0.198 / 4.39	+0.204 / 4.79
GO multi-hot with the minimum-norm flux appended	6539	+0.068 / 2.20	+0.208 / 4.59	+0.173 / 4.19
measured Mülleder amino-acid pool of the strain (19 mM columns)	19	+0.149 / 3.99	+0.285 / 4.19	+0.280 / 4.79
GO multi-hot with the measured pool appended	2381	+0.102 / 2.20	+0.174 / 4.59	+0.264 / 5.99
HiGHS parsimonious store with GO and the measured pool appended	6558	+0.156 / 3.19	+0.195 / 4.19	+0.247 / 6.19
measured pool with the HiGHS parsimonious vector appended, on the pool's genes	4196	+0.095 / 1.80	+0.153 / 2.60	+0.220 / 3.99
measured Cooper 2010 amino-acid peaks of the strain (17 relative columns)	17	-0.001 / 0.60	+0.094 / 1.80	+0.113 / 1.20
ProtT5 (027's ladder)	1024	+0.082 / 3.00	+0.144 / 4.59	+0.143 / 4.39
GLPK store with ProtT5	9379	+0.044 / 2.00	+0.066 / 2.00	+0.121 / 4.99

4 Where the module stands on this target ✗

Six things are measured. The whole-vector store affine lifts the width-12 backbone's rank to where the width-90 backbone sits and, on the strain as built, lifts its call too; the two do not add, at either width. The gain lives in the production half of the vector and in a free affine that trains at full rate; it does not come from the first epochs, and a release rule on the validation curve keeps it where a fixed late start loses it, while a nonlinear head over the store loses it and brings the plateau back. The physically stated stores read in the transformer sit at the parsimonious store's level: the adjustment store within the seed spread with half its spread and the best single-seed call, the clean parsimonious vector a little above it on five seeds of six with the best width-12 ensemble, and the driving-force concentrations at the same rank with twice the spread; the second law adds a second representation at the flux store's level, not a higher one. Without any backbone the clean vector read by nearest neighbors carries more rank than any single transformer arm, and the sampled driving-force polytope is the one flux-side representation every model reads above 0.12; in the transformer the polytope's mean reads at the flux store's level on every seed above the control, with the best width-12 ensemble rank, and the sampled cone is the weakest flux representation. The release rule that lifted the degenerate GLPK store adds nothing to the adjustment store and takes from the HiGHS store (rounds 149, 152); a store without the vertex noise is read best from the first epoch. The annotation family sets the level: a perceptron over the propagated GO multi-hot reads 0.200 in three seconds a seed, above every transformer arm, and the same columns in the affine slot read 0.202 with the seeds identical (round 150), the backbone adding nothing; the HiGHS vector appended to GO in that affine reads 0.211, above GO alone on five seeds of six, the highest rank of the study (round 151). Dropout 0.3 on the store arm reads 0.145, the best single-seed rank and call of any store arm, where width 6 to 90 left the level flat (144, 145). Every arm overfits by epoch 60 and averaging seeds is worth more than any representation on the call. On the same split the comparators sit below the best arm, the closest by 0.008: a random forest over GO with the HiGHS vector, the best arm's own input, 0.204 with a top-50 of 4.79, cosine nearest neighbors over the HiGHS store 0.195, the ridge 0.152, nearest neighbors over ProtT5 0.145, the Flux Cone Learning random forest 0.039 (Tables 14 and 8); the best call of any genotype-side read is the forest over GO alone, 5.39. The strain's own measured amino-acid pool, nineteen columns, reads 0.285 by nearest neighbors and 0.280 by forest on the same split, above every genotype-side read and every transformer arm, and appended to GO with the HiGHS vector gives the forest the best call of any read, 6.19 (Section 3); the transformer reading the same columns is queued (175, 176). The amino-acid endpoint shares strain-level signal with this one: a deletion's measured amino-acid pool, Mülleder's nineteen concentrations through a ridge, reads its betaxanthin level out of fold at 0.226 over 4,477 strains, and appended to the gene's GO multi-hot in the same ridge adds 0.072 to what the annotation reads alone (0.165 to 0.237), the sibling document's Section 4 (`notes-tex/032-amino-acid-module`). The three levers on one cell, the HiGHS store under the rule with dropout 0.3, read 0.125, under the GLPK store at dropout 0.3 alone (round 154): the release is the lever that costs on the clean

store. The dropout curve turns between 0.3 and 0.5: at 0.5 three seeds of six never leave the plateau and the arm falls back to the incumbent's level, 0.125 against 0.121 (round 155), so 0.3 is a maximum, not a direction. The polytope's spread beside its mean adds nothing on single seeds and gives the highest six-seed rank of any flux-side width-12 arm, 0.155 (round 156): the region's value is in the ensemble. The GO affine under the release rule reads 0.199, the backbone escaping first adding nothing to the annotation's single read; the rule buys seed diversity, worth 0.003 on the ensemble (0.206, round 158); dropout 0.3 on the same arm moves it by less than a thousandth (round 159), so the annotation arm is the affine alone and no setting of the transformer around it has moved it; the same holds for the flux-and-annotation affine, which dropout 0.3 leaves at 0.212 with the flux gain beside GO reproduced at 0.010 (161). The polytope's mean appended to that affine takes 0.013 from it on five seeds of six and the stack reads under GO alone (163): the metabolite side does not add to the annotation the way the flux vector does. On twelve seeds the best arm holds at 0.214 with a standard error of 0.002, 0.011 above the forest over its own columns and 0.019 above the nearest-neighbor read of the HiGHS store, with a twelve-seed ensemble of 0.216 (round 164). Adding the region's width to that stack costs again, and the cost grows with the columns added (0.193, round 166): the metabolite side does not enter this affine's read at any width. The adjustment store in place of the parsimonious one adds 0.006 to the annotation on every seed against the parsimonious vector's 0.009 (round 167): both flux programs enter the annotation's read where the metabolite programs do not. Weight decay on the affine's own weights runs 0.211, 0.210, 0.204 from none to 0.1 to 1.0 (rounds 169, 170), the heavy setting selecting fifty epochs later with twice the seed spread: no setting of the transformer and no regularization of the affine has moved the annotation arms. The two flux programs together beside GO read 0.216, above the parsimonious vector with GO on five seeds of six at a paired spread of 0.003 (round 171), the highest single-seed rank of the study and the second time the flux side has entered the annotation's read; the call does not follow it. On twelve seeds the annotation-only arm reads 0.203 with a standard error of 0.0005, the study's most reproducible cell, and the flux vector's gain beside it is +0.012 on eleven of twelve (round 172). Against the affine-free control on the same twelve seeds the annotation arm is +0.127 on twelve of twelve, and its ensemble, 0.203, stays twice the control's twelve-seed ensemble of 0.102 (round 173): unlike the amino-acid target, where averaging the control's seed noise passed the annotation's small gain, here the gain is five times what the averaging recovers. The measured pool in the affine adds 0.006 to GO on every seed and nothing beside the HiGHS vector (rounds 175, 176), a linear read of columns whose signal the ladder's nearest neighbors carry at 0.285: the slot bounds the read, not the data. On twelve seeds the two flux programs beside GO read 0.217 against the parsimonious vector's 0.214, +0.003 within a paired spread of 0.007 (round 177), so round 171's gain does not hold at its first size and the two arms share one level; the three families in one affine read 0.218 (round 178), the pool adding nothing beside the flux programs. Every linear combination of the annotation with the flux and measured sides reads 0.208 to 0.218: the affine's ceiling on this input family. The pool alone in the affine reads 0.195 with the widest seed spread and the lowest call of any store arm (round 179), above the ridge on the same columns (0.149) and 0.09 under their nearest-neighbor read: the ceiling is the slot's. The same columns through a hidden layer of 64 read 0.240, +0.045 over the linear slot on five of six and the highest single-seed mean of any transformer arm, with a six-seed ensemble of 0.292 that passes the ladder's nearest neighbors (round 180): the pool's nonlinear signal does reach the transformer once the slot is nonlinear; a 400-epoch budget selects the same epochs and reads the same values on every seed (190), so 0.240 is the arm's level, not a bound. The minimum-norm flux store, the unique point with no vertex noise, reads under the sparse HiGHS vertex both alone (0.093 against 0.130, round 181) and beside GO (0.204 against 0.211 on six of six, round 182): the flux side's gain beside the annotation lives in the sparse solution. The annotation appended to the pool in the hidden layer takes the read back to the annotation's level, 0.206 with selection at epochs 2 to 4 (185): with GO in the layer the arm is the GO affine, and the pool's gain is gone, as the ladder's ridge and nearest neighbors lost it when the GO columns joined. On twelve seeds the pool through the layer holds at 0.254 with a standard error of 0.012, +0.051 over GO alone on eleven of twelve, with a twelve-seed ensemble of 0.297 (186): the highest level and the highest ensemble of the study, above every ladder read of the same columns, and the call under the annotation's throughout. The layer's width matters: through 256 the pool reads 0.279 with half the seed spread, +0.039 over 64 on five of six and +0.084 over the linear slot on six of six, the single-seed level at the ladder's nearest-neighbor read and the ensemble (0.288) above it (187). Every other family appended to the pool inside the layer costs it: the annotation (185, −0.034), the three families (188, back to the annotation's 0.217) and the flux vector (191, −0.054, under both parents). The pool's nonlinear read is best taken alone, and the head that reads it beside the other families through separate slots is the arm not yet built. Without the transformer the same columns give a perceptron 0.236 at width 1,024 and 0.240 with GO appended, the annotation adding there at every width (Section 3): the transformer's hidden slot reads the pool 0.04 above any standalone perceptron, and what the annotation does to the pool's read depends on the head it enters. The perceptron with two slots, GO linear beside the pool's hidden layer, reads 0.265, above the shared layer (0.240) and the pool alone (0.236) at every width, and holds under depth where the pool alone falls to 0.19: the annotation adds to the pool when the two do not share a hidden unit. That two-slot head inside the transformer is the arm in flight. The width curve of the pool's hidden read is 0.240, 0.262, 0.279 and 0.243 at 64, 128, 256 and 512 on the same seeds (192, 194): a maximum at 256, which twelve seeds hold at 0.278 with a standard error of 0.007, +0.075 over GO alone on twelve of twelve, with a twelve-seed ensemble of 0.296 (193). In the transformer the two-slot head reads 0.224 at 64, 0.225 at 128 and 0.225 at 256 (197, 200, 198), the width of the pool's slot never read, above every cell with the annotation in it on every seed with the tightest band and the best single-seed call (top-50 5.66), and 0.054 under the pool alone through 256: it selects at epochs 5 to 15 on the annotation's early peak, before the pool's slot has trained,

where the perceptron's two slots selected at 64 to 165. The linear slot at a tenth and a hundredth of the rate is the fix in flight (201, 202). With the three families in the two slots, GO and both flux programs linear beside the pool through 256, the head reads 0.239 with a band of 0.003 (199): the flux programs add 0.014 to the annotation's slot on every seed, and this is the best arm of the study with the annotation in it, 0.04 under the pool alone. A 400-epoch budget leaves the pool through 256 where 200 epochs left it, the same epoch and the same value on every seed (196), so the width curve's maximum is a level. Cells queued: two slots at 128 (200), dropout 0.3 and the standardized encoding on the pool through 256 (203, 204), on the amino acids the score through a hidden layer of 256 with Cooper's peaks linear and through the same layer (205 to 207), and Cooper's peaks alone through 256 (211), a substitute for the pool rather than an addition, where the perceptron reads the peaks alone at 0.131 and appended to the pool as steadying it (0.245 against 0.219 at width 256, the spread from 0.042 to 0.010) (Section 3). Dropout 0.3 inside the transformer leaves the pool through 256 at 0.279 with a paired difference of zero (203), and standardizing the pool instead of $\log(1 + v)$ costs it 0.10 (204). What the transformer contributes is now measured (Section 3.1): on the pool through 256 the backbone's head is a constant on two seeds of six and adds 0.05 on the other four, the slot alone reads 0.257 and the arm 0.279; with the annotation in the slot the head is a constant on every seed and the read is the slot's. No cell of the study has read the graphs, since the regularizer and the hard mask index a second block and every cell ran one. The layer-pattern ladder at width 36 with nine heads (212 to 216) and the release rule on the pool slot (210) are queued.